Topology Simplification of District Heating Networks via Artificial Intelligence-based Node Aggregation

Dubon
RODRIGUE

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Invited:	Dr. Jérôme Henri Kämpf		

Defended on 23 October 2025 in Nantes (France)

District Heating Network (DHN)

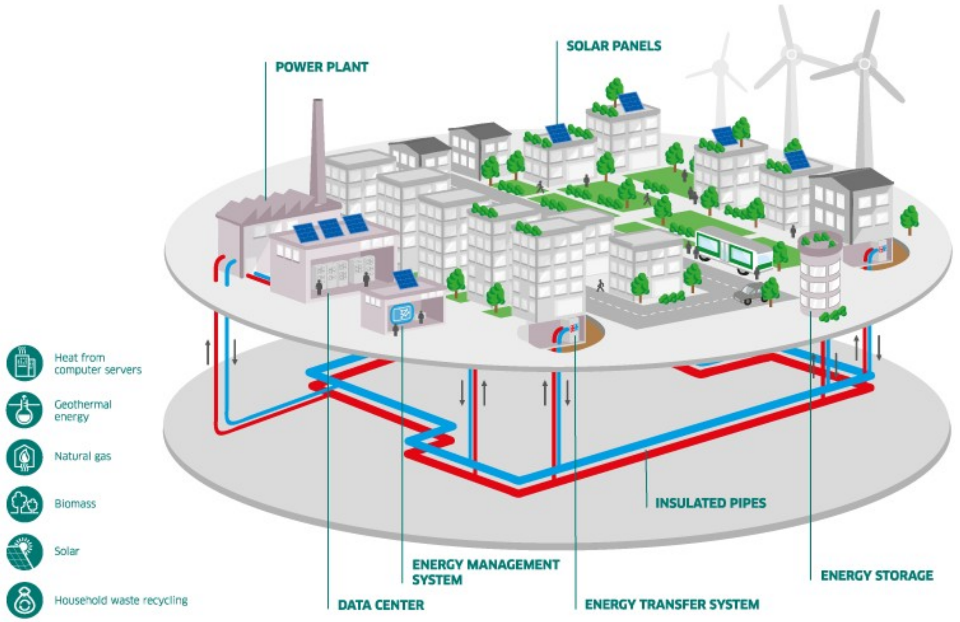


Illustration of District Heating Network (DHN)
Source: ENGIE Inc.

- System to distribute heat in localized areas

- Main components:
  - Producers (central or distributed)
  - Consumer substations
  - Network of distribution pipes (supply / return)
  - Heat-carrier fluid (*e.g.*, heated water)

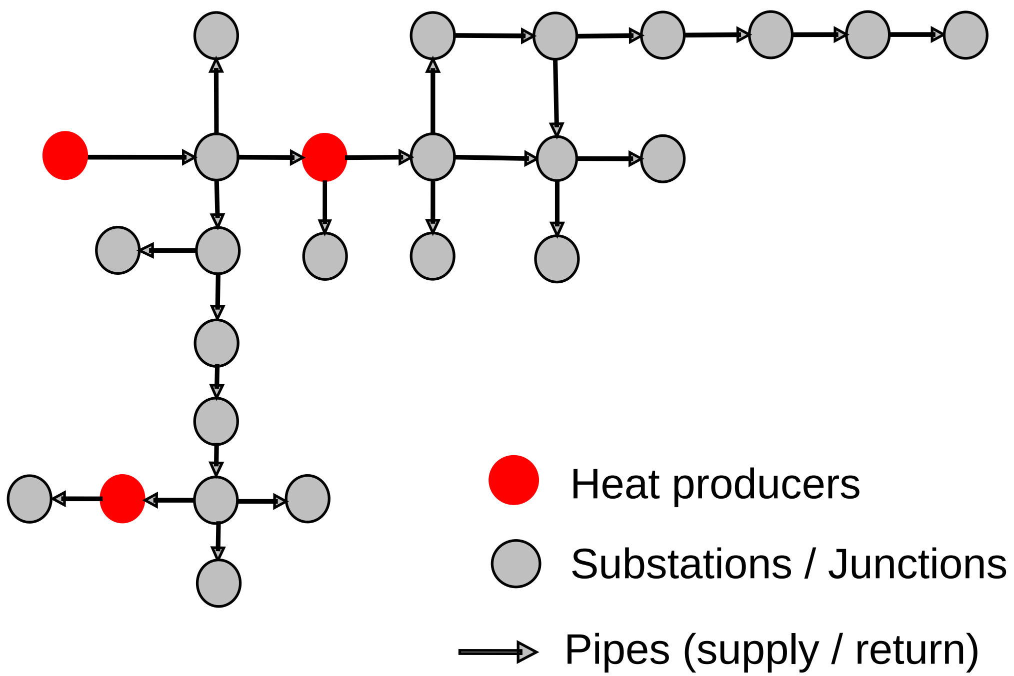
- Opportunities to **reduce GHG emissions** in heat supply (*e.g.*, integration of renewable or optimized distribution)

Operational Objectives of DHN

	Time scale	**Operational Objectives**
>10 years – 1 year	Long-term	Planning / Sizing (e.g., renewable integration)
1 year – 1 hour	Mid-term	Operations planning
1 hour – 1 minute	Short-term	Optimal control

- Multiple simulations to evaluate several scenarios
- High details captured by dynamic simulation model

Dynamic Simulation Challenges



$\mathcal{V}$: sets of nodes
$\mathcal{E}$: sets of edges

Graph formulation of DHN using node-method
(*e.g.*, HeatGrid)

Dynamic Simulation Challenges

Solve a system of complex physical equations

$\mathcal{F}_{sim}(\mathcal{V}, \mathcal{E})$

- Mass / Enthalpy conservation
- Heat Transport Equation
- Pressure drops

Computational **costs** of $\mathcal{F}_{sim} \propto (|\mathcal{V}| + |\mathcal{E}| \times N_{vol})$

$\mathcal{V}$: sets of nodes

$\mathcal{E}$: sets of edges

N_{vol} : number of finite volumes

Existing Strategies for this Challenge

Simplifying physical equations

$$\mathcal{F}_{sim} \simeq f$$

- Approximative physical equations [Hoy+18]
- Model Order Reduction (MOR) [Gue+16]
- Machine Learning (ML) [Bou+14]

✓ Accuracy
✓ High acceleration
× Flexibility

Spatial simplification

$$\mathcal{F}_{sim}(v, e)$$

$$|v| < |\mathcal{V}|, |e| < |\mathcal{E}|$$

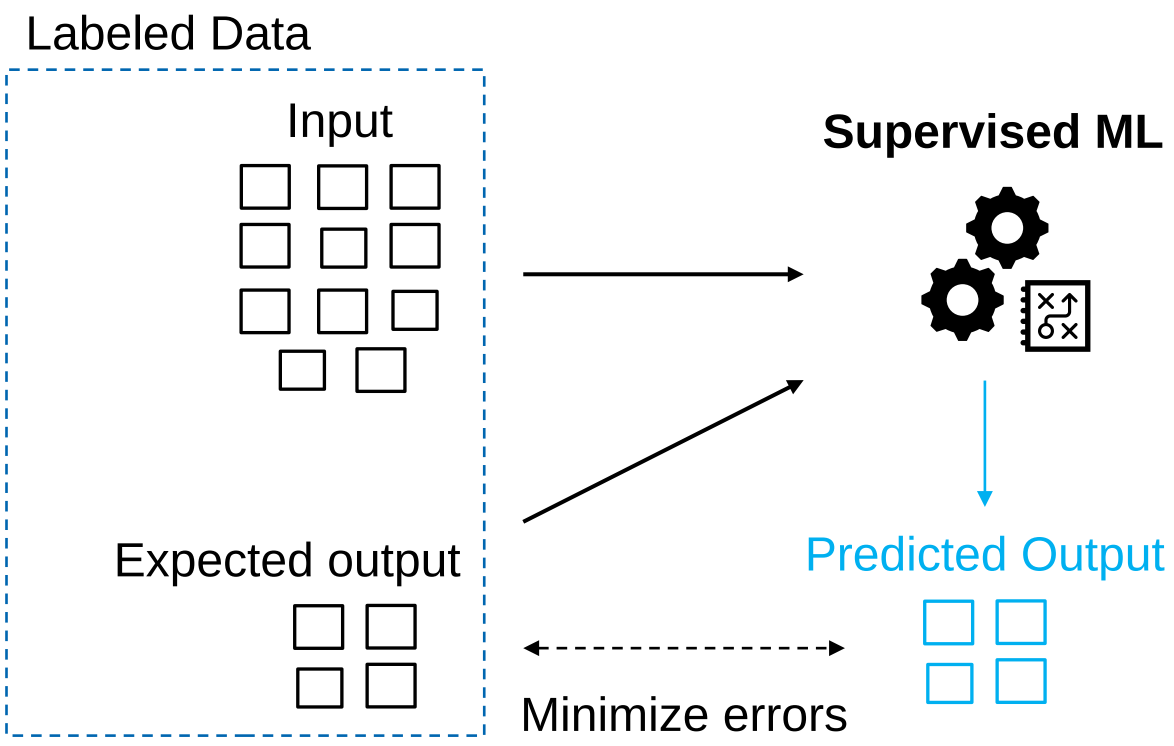
- Nodes aggregation [SXM24]
- Topology simplification [Loe+01]

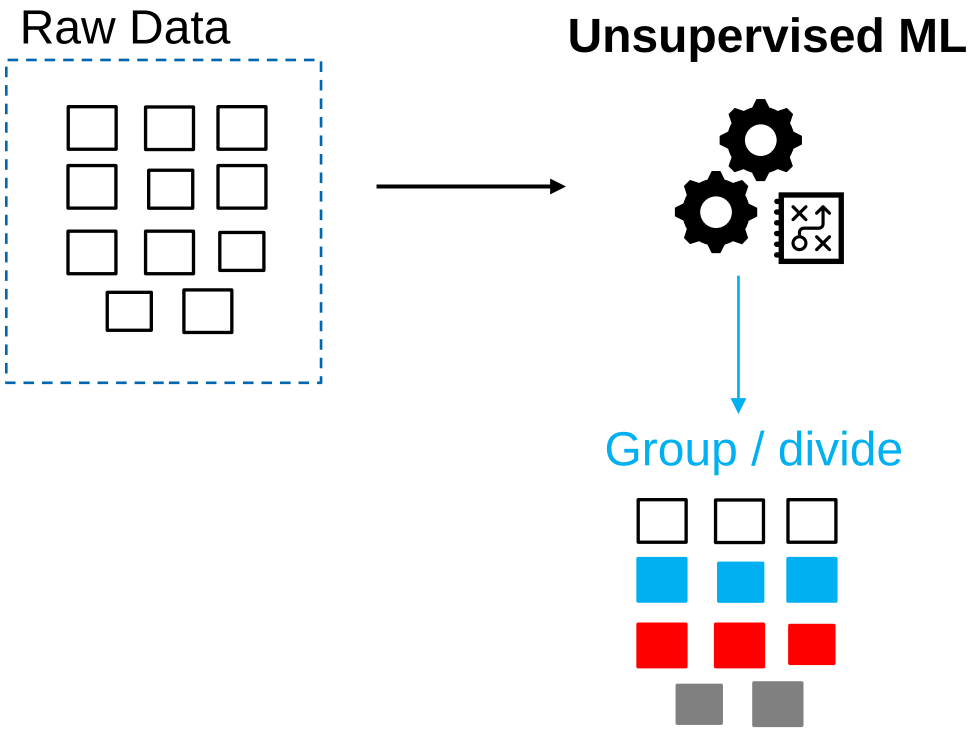
✓ Interpretability
✓ Flexibility
× Representativity

Our approach
(ML + Topology simplification)

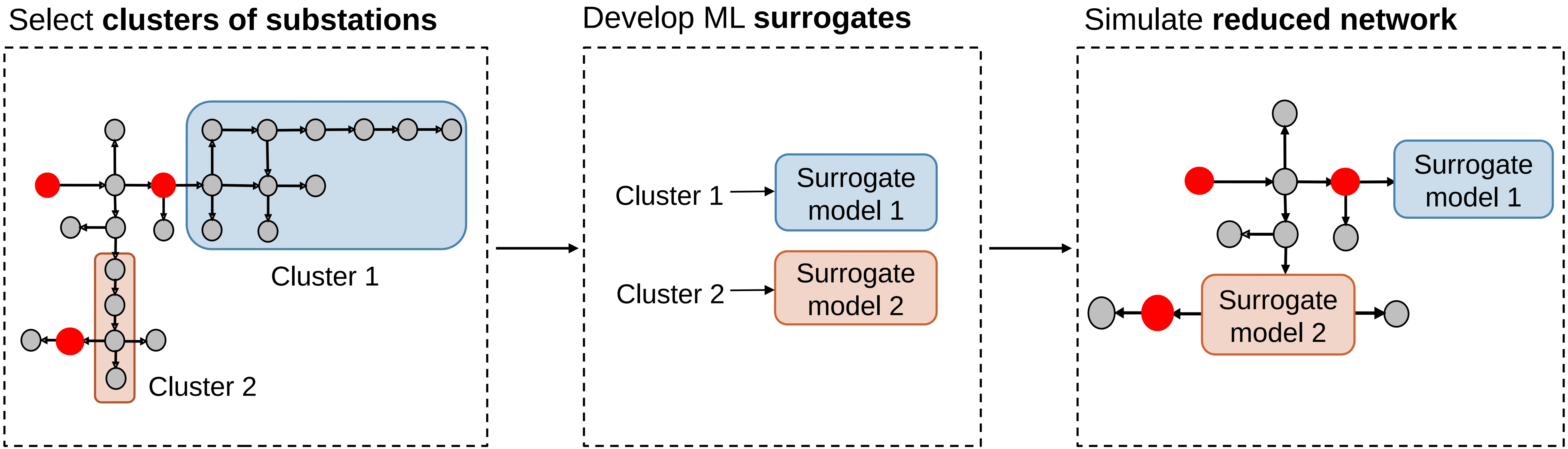
Artificial Intelligence (AI)

Machine Learning (ML) is the core of modern AI systems

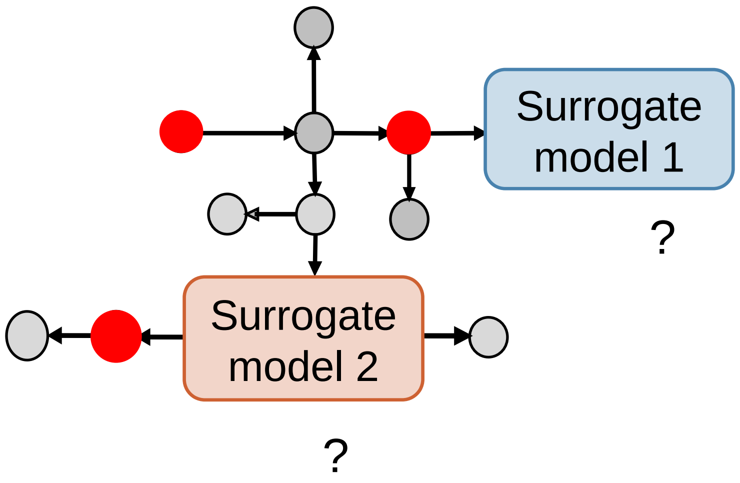




Thesis General Objectives



Research Questions



Research Question Q1 - What ML model architectures, features, and training methodologies enable learning and preserving the physical dynamics of substation clusters within DHNs?

Research Question Q2 - What properties of substation clusters and unsupervised ML methodologies enable the automatic identification of the optimal clusters that support high accuracy?

Research Question Q3 - What strategies enable efficient integration of surrogate ML models into DHNs' dynamic simulations while reducing computational costs and preserving physical accuracy?

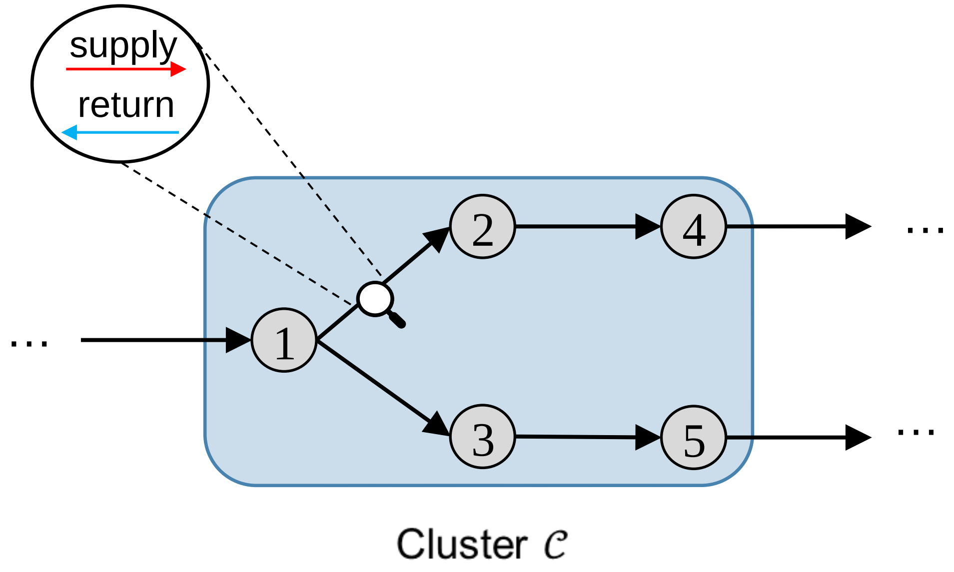
Research Question Q1 - What ML model architectures, features, and training methodologies enable learning and preserving the physical dynamics of substation clusters within DHNs?

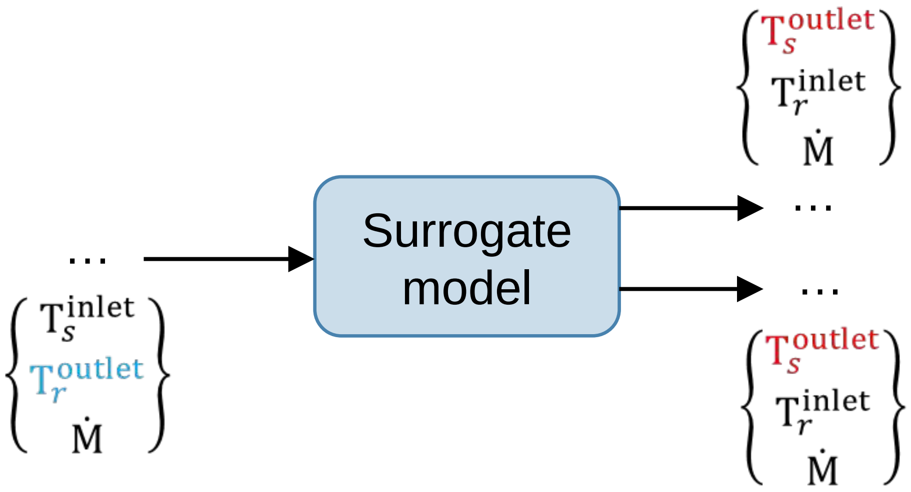
Q1.1 – What are the Input and Output features of the ML Surrogate Models?

Q1.2 – Which data and training procedures can be used to train the surrogate models?

Q1.3 – Which ML architectures yield best compromise in accuracy and computational efficiency?

Q1.1 – What are the Input and Output features of the ML Surrogate Models?





Output features: Outgoing water temperatures

Input features: Incoming water temperatures
Incoming water mass flow rates
Composing nodes heating demands profiles

Q1.2 – Which data and training procedures can be used to train the surrogate models?

- Training data from physical simulation (*HeatGrid*)
  - January to Mid-June, heating law production temperature
  - 80 % used for training/validation and 20% for testing

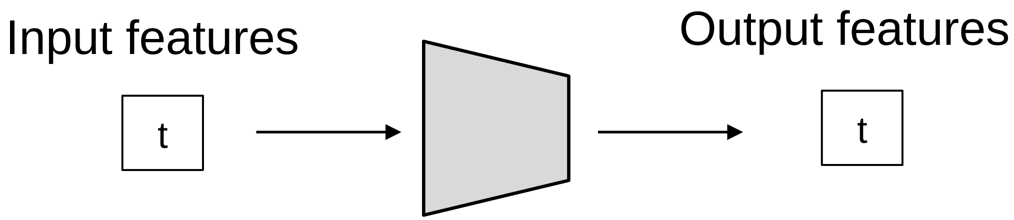
- Supervised ML, trained to minimize **Mean Absolute Error** (MAE)

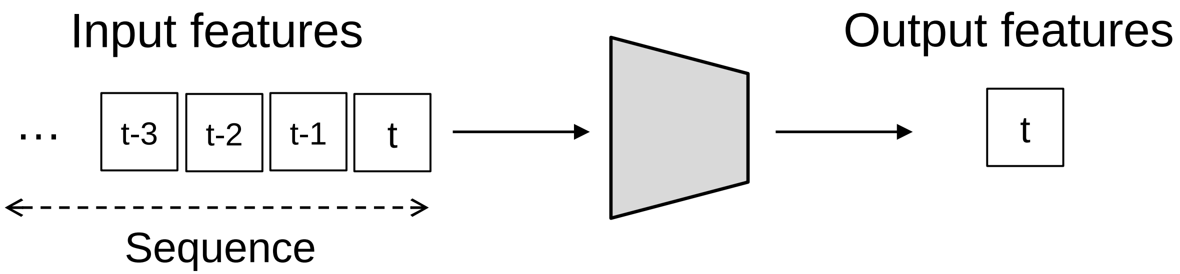
$$\text{MAE (°C)} = \frac{1}{N_t N_e} \sum_{e=1}^{N_e} \left(\sum_{t=1}^{N_t} |y_e^t - \tilde{y}_e^t| \right)$$

y: Physical temperatures

$\tilde{y}$: Predicted temperatures

Q1.3 – Which ML architectures yield best compromise in accuracy and computational efficiency?



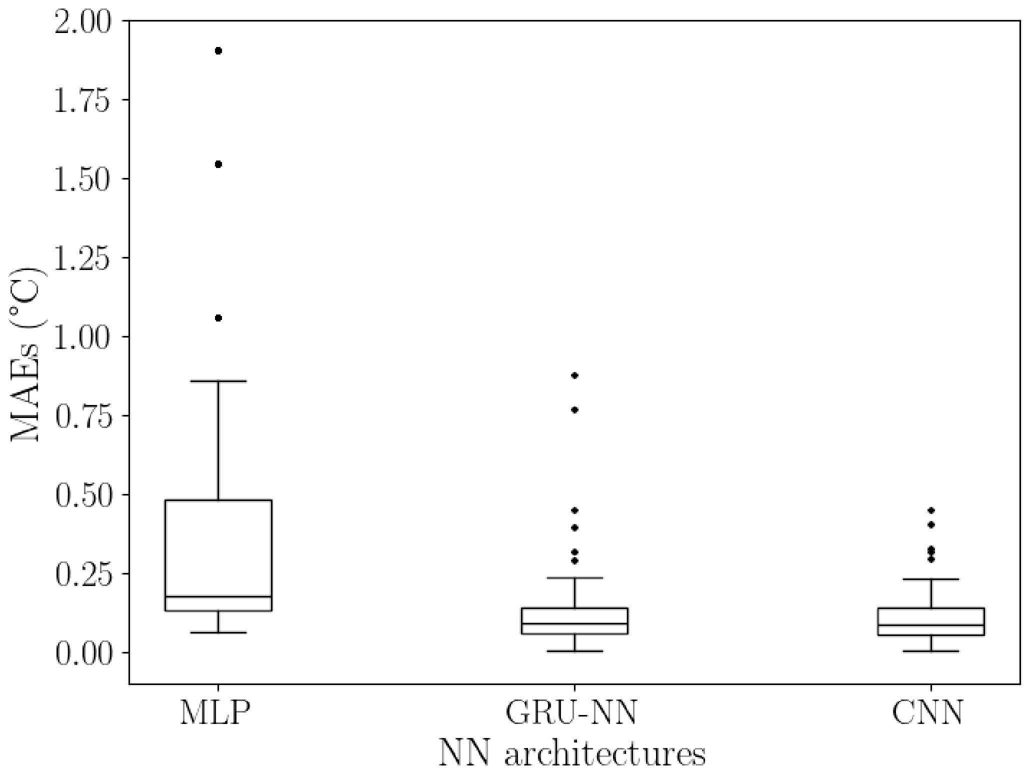


- **Single-step input ML models**
  - Takes single time step of input features
  - Multilayer Perceptron (MLP)

- **Sequential input ML models**
  - Takes temporal sequence of input features
  - Recurrent Neural Network using GRU (GRU-NN)
  - Convolutional Neural Network using Conv1D (CNN)

Q1.3 – Which ML architectures yield best compromise in accuracy and computational efficiency?

Learned **220** substation clusters selected from **4** distinct case study DHNs



	avg. MAE (°C)	avg parameters
MLP	0.797 (±0.21)	7485
GRU-NN	0.172 (±0.07)	6635
CNN	0.115 (±0.02)	8425

- GRU-NN proposes best balance between complexity (linked to inference time) and accuracy

Research Question Q2 - What properties of substation clusters and unsupervised ML methodologies enable the automatic identification of the optimal clusters that support high accuracy?

Q2.1 – Which metrics define the quality of a substation cluster?

Q2.2 – Which properties of a cluster most influence its quality?

Q2.3 – What clustering approach can be used to identify high-quality clusters?

Q2.1 – Which metrics define the quality of a substation cluster?

3 **quality metrics** quantifying the quality of a clusters

MAE (°C)
Energy Error (MWh)
Energy Error relative to loss (%)

A high quality cluster: a cluster of substations which lead to accurate surrogate model

Q2.2 – Which properties of a cluster most influence its quality?

$$\begin{pmatrix} \text{MAE (°C)} \\ \text{Energy Error (MWh)} \\ \text{Energy Error relative to loss (\%)} \end{pmatrix} = f\begin{pmatrix} \text{Topological properties} \\ \text{Physical properties} \end{pmatrix}$$

Q2.2 – Which properties of a cluster most influence its quality?

13 Topological properties considered

Size

Internal density

Internal out-degree (avg, max)

Internal in-degree (avg, max)

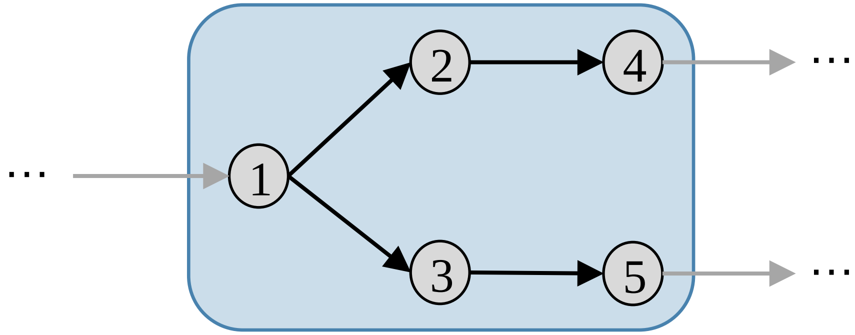
Cut size (# external edges)

External in-degree (avg, max)

External out-degree (avg, max)

Cut ratio

Conductance



Cluster $\mathcal{C}$

e.g., size $(\mathcal{C}) = 5$

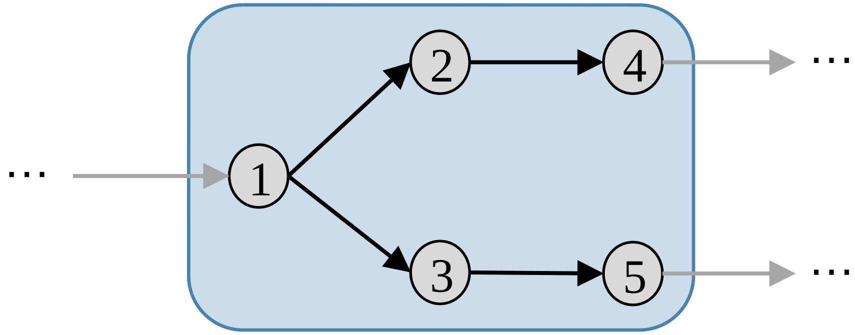
Q2.2 – Which properties of a cluster most influence its quality?

13 Topological properties considered

Size
Internal density
Internal out-degree (avg, max)
Internal in-degree (avg, max)

Cut size (# external edges)
External in-degree (avg, max)
External out-degree (avg, max)

Cut ratio
Conductance



Cluster $\mathcal{C}$

$$\text{internal density} = \frac{\#\text{ internal edges}}{\text{size}\,(\text{size} - 1)}$$

Q2.2 – Which properties of a cluster most influence its quality?

13 Topological properties considered

Size

Internal density

Internal out-degree (avg, max)

Internal in-degree (avg, max)

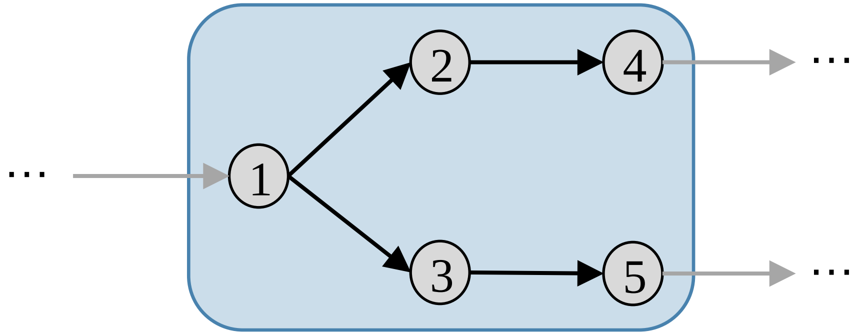
Cut size (# external edges)

External in-degree (avg, max)

External out-degree (avg, max)

Cut ratio

Conductance



Cluster $\mathcal{C}$

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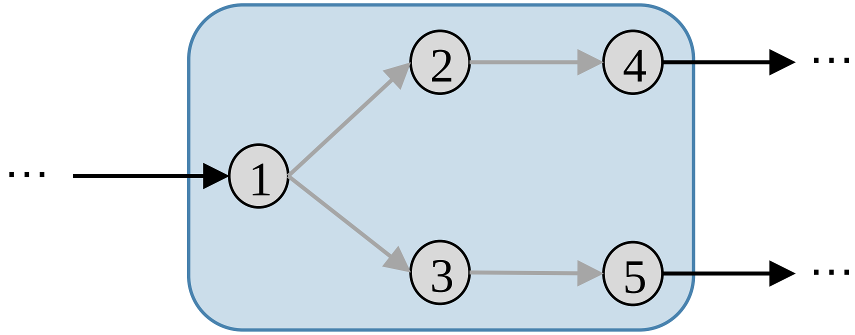


Cluster $\mathcal{C}$

e.g., cut size $(\mathcal{C}) = 3$

Q2.2 – Which properties of a cluster most influence its quality?

<u>13 Topological properties considered</u>

Size

Internal density

Internal out-degree (avg, max)

Internal in-degree (avg, max)

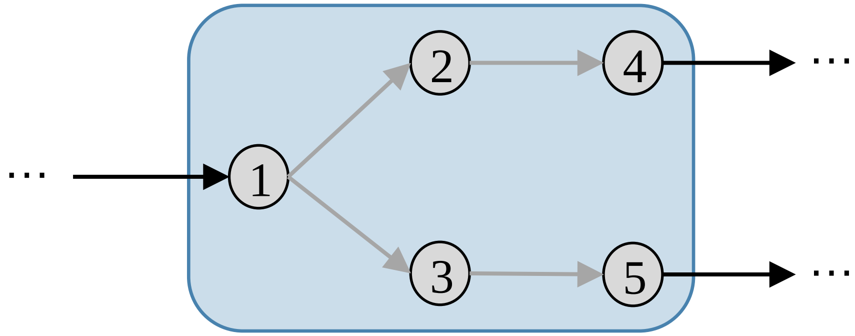
Cut size (# external edges)

External in-degree (avg, max)

External out-degree (avg, max)

Cut ratio

Conductance



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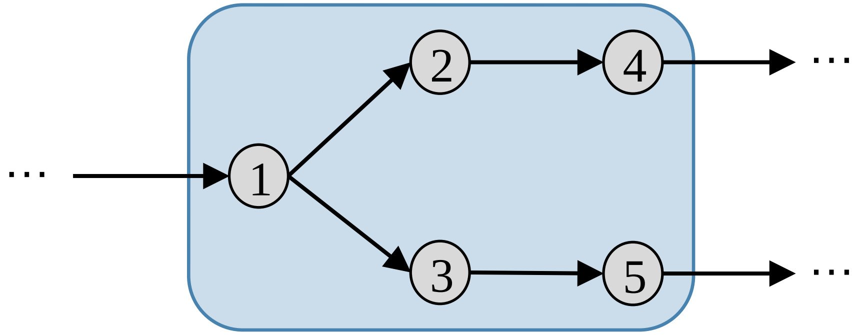
Cut ratio

Conductance



Cluster $\mathcal{C}$

$$\text{cut ratio} = \frac{\text{cut size}}{\text{size} * \text{outside nodes}}$$

$$\text{conductance} = \frac{\text{cut size}}{\text{internal density}}$$

Q2.2 – Which properties of a cluster most influence its quality?

8 Physical properties considered

Geographical max. spatial extent

Internal pipes avg. length

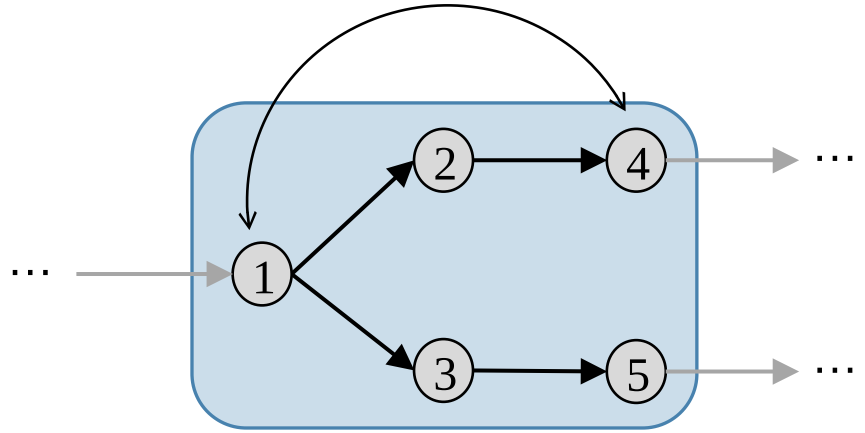
Internal pipes avg. diameter

Internal pipes avg. flow velocity

Transport thermal loss

Transport delay time (avg, max)

Heating consumption avg. difference



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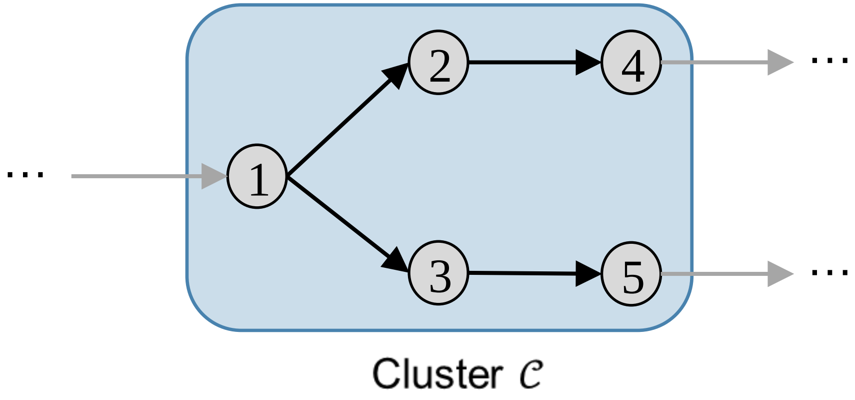

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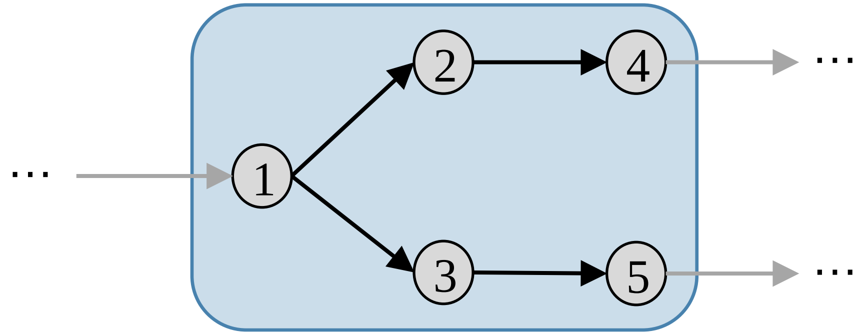

Cluster $\mathcal{C}$

Transport thermal loss = Incoming thermal energy
– Consumed thermal energy
– Outgoing thermal energy

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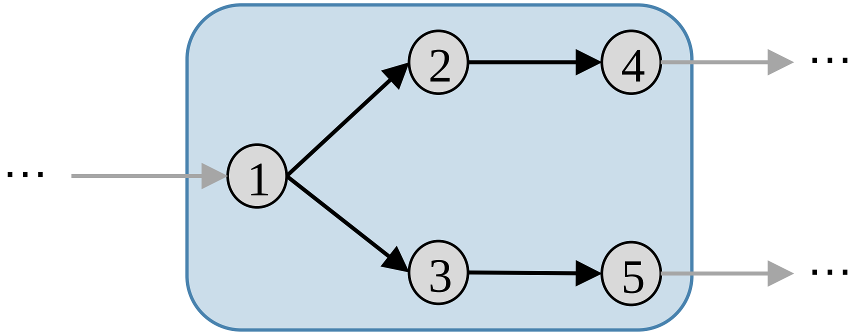

Cluster $\mathcal{C}$

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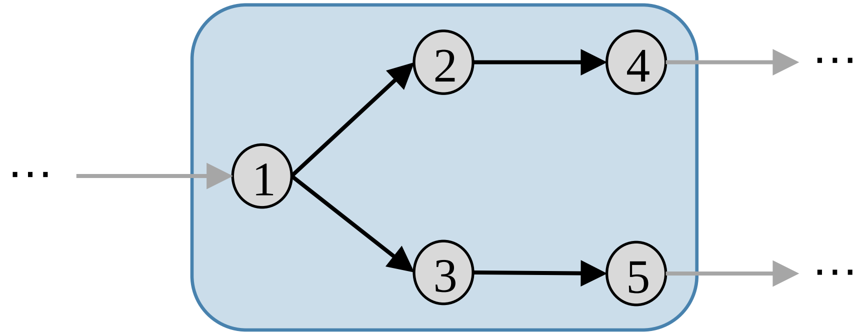


Cluster $\mathcal{C}$

Q2.2 – Which properties of a cluster most influence its quality?

Multi-variate correlation (Spearman) analysis on **2694** trained substation clusters from **17** different case study DHNs

	Cut size	Transport thermal loss	Transport delay time (avg)
MAE	0.081	0.22	0.41
Energy Error	0.56	0.54	-0.21
Energy Error relative to loss	0.46	-0.13	-0.38

High quality:

- low cut size,
- low thermal loss
- short transport delay time

Q2.3 – What clustering approach can be used to identify high-quality clusters?

For all pair of nodes $\{u, v\} \in \mathcal{V}$

$$\mathrm{dist}_{(u,v)} = \begin{cases} \alpha\, \overline{\mathrm{dist}}^{\mathrm{tp}}_{(u,v)} + \beta\, \overline{\mathrm{dist}}^{\mathrm{tm}}_{(u,v)} + (1-\alpha-\beta)\, \overline{\mathrm{dist}}^{\mathrm{tr}}_{(u,v)} & \text{if } u \text{ and } v \text{ connected} \\ 0 & \text{if } u = v \\ 1 & \text{else} \end{cases}$$

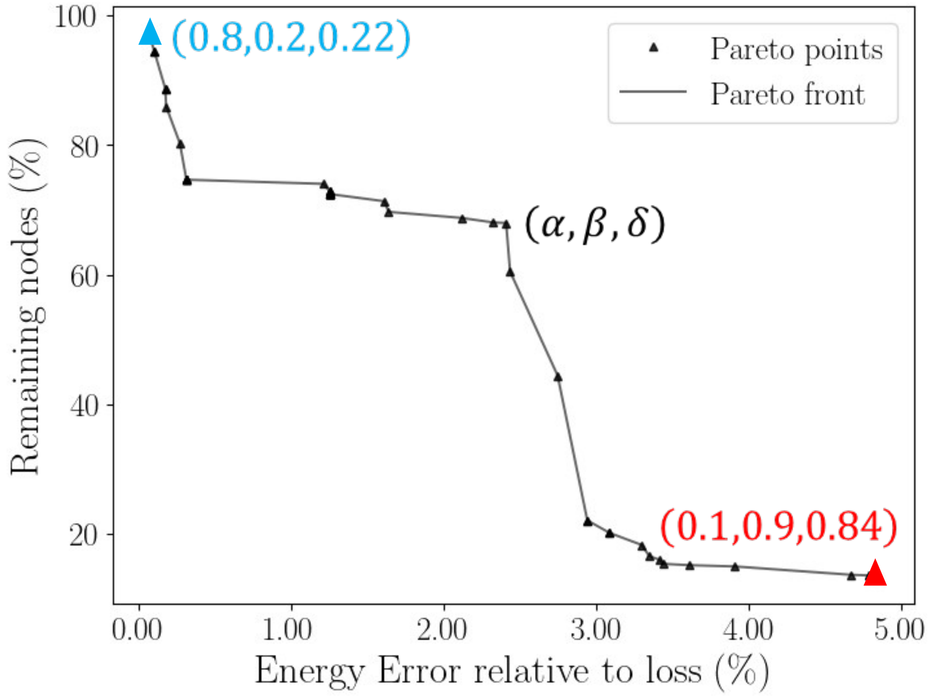
$\overline{\mathrm{dist}}^{\mathrm{tp}}$: cut size (normalized) $\overline{\mathrm{dist}}^{\mathrm{tm}}$: thermal transport loss (normalized) $\overline{\mathrm{dist}}^{\mathrm{tr}}$: transport delay time (normalized)

Leveraged in a Hierarchical Agglomerative Clustering (distance threshold $\boldsymbol{\delta}$, single linkage)

Decide the hyperparameters (α, β, δ) ?

Q2.3 – What clustering approach can be used to identify high-quality clusters?

Performed clustering for 16 DHNs with all hyperparameters combination



- Trade-off between clusters' quality and obtained spatial reduction

- Hyperparameters can incorporate **user-strategy preferences**

i.e., High quality clusters

i.e., High spatial reduction

Research Question Q3 - What strategies enable efficient integration of surrogate ML models into DHNs' dynamic simulations while reducing computational costs and preserving physical accuracy?

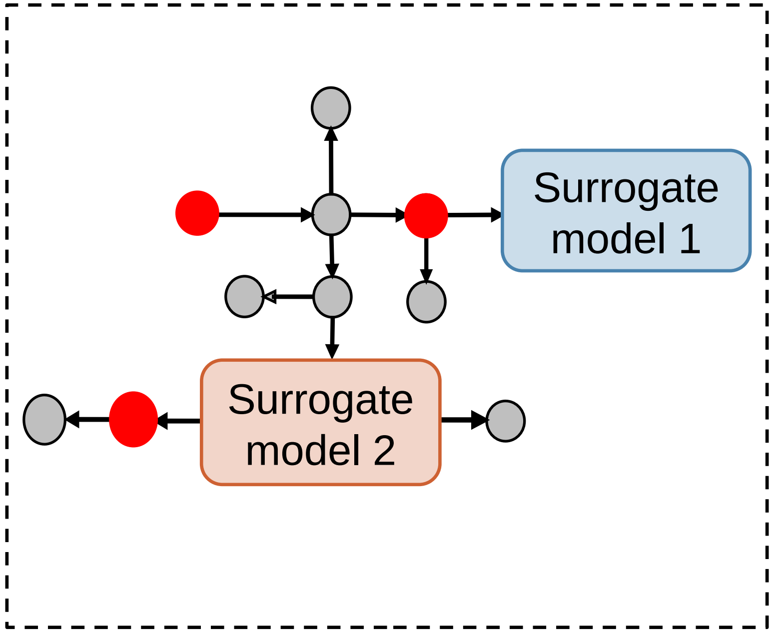
Q3.1 – How to integrate surrogate models within the simulation of the reduced network?

Q3.2 – What are the performances of this new simulation framework?

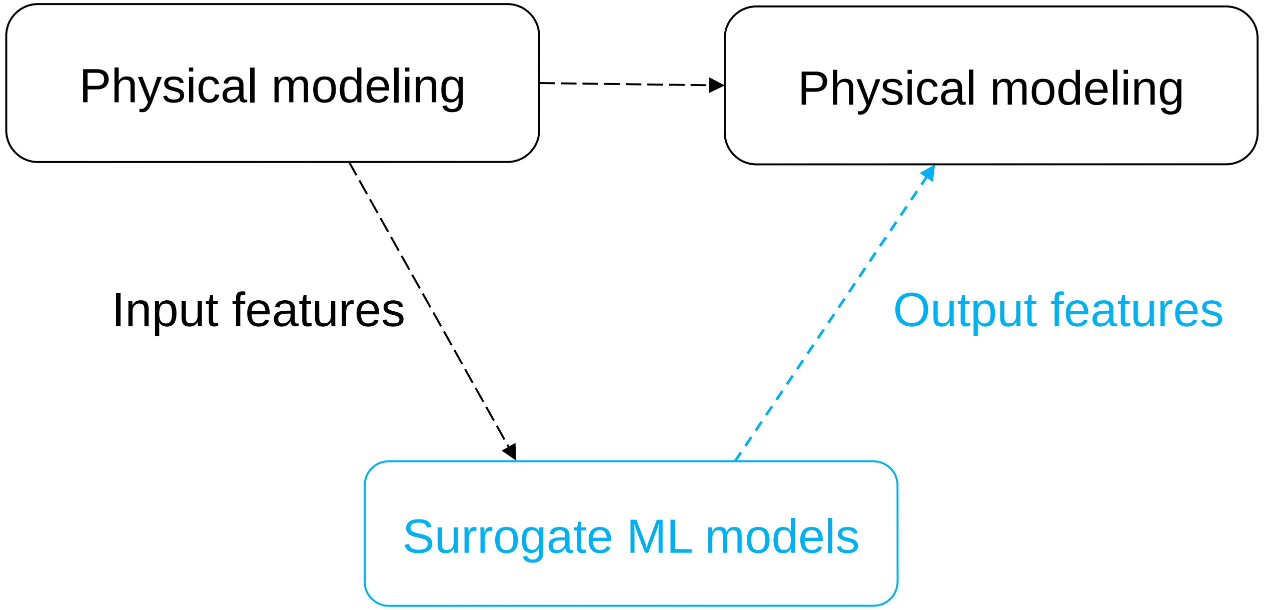
Q3.3 – Which strategies allow high accuracy and high computational reduction of this new simulation framework?

Q3.1 – How to integrate surrogate models within the simulation of the reduced network?

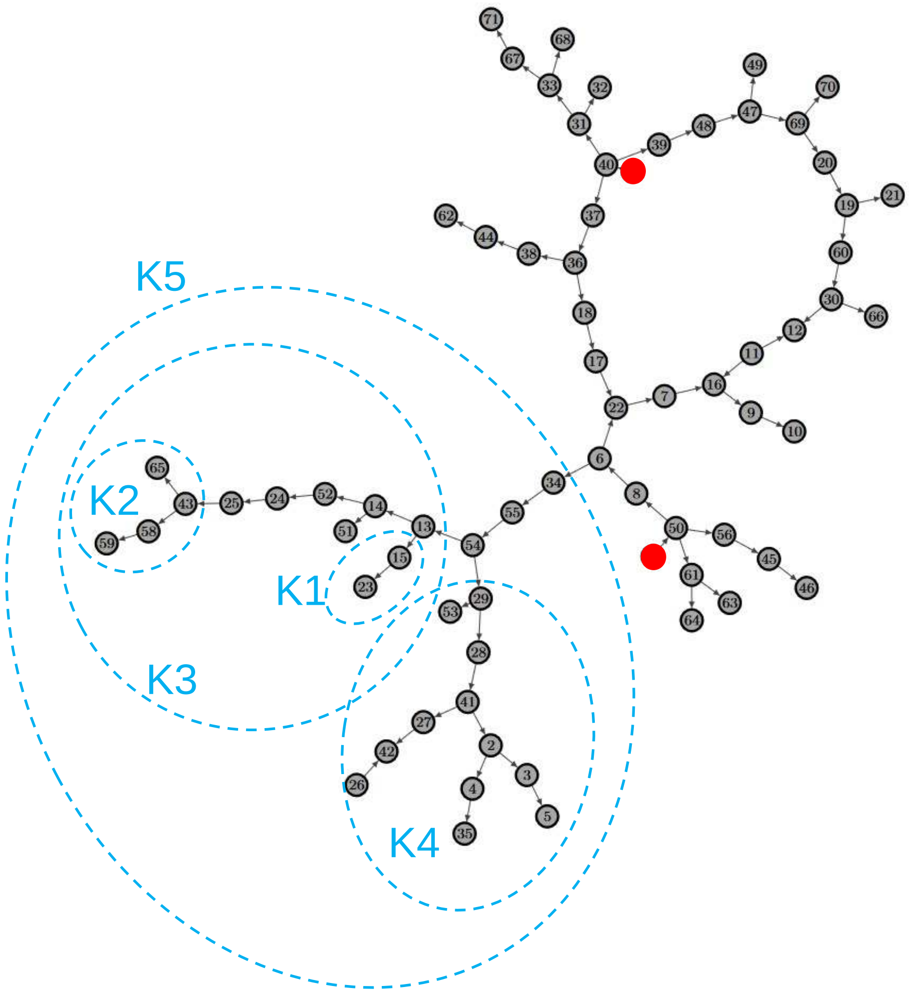
Simulate **reduced network**



Hybrid Simulation framework

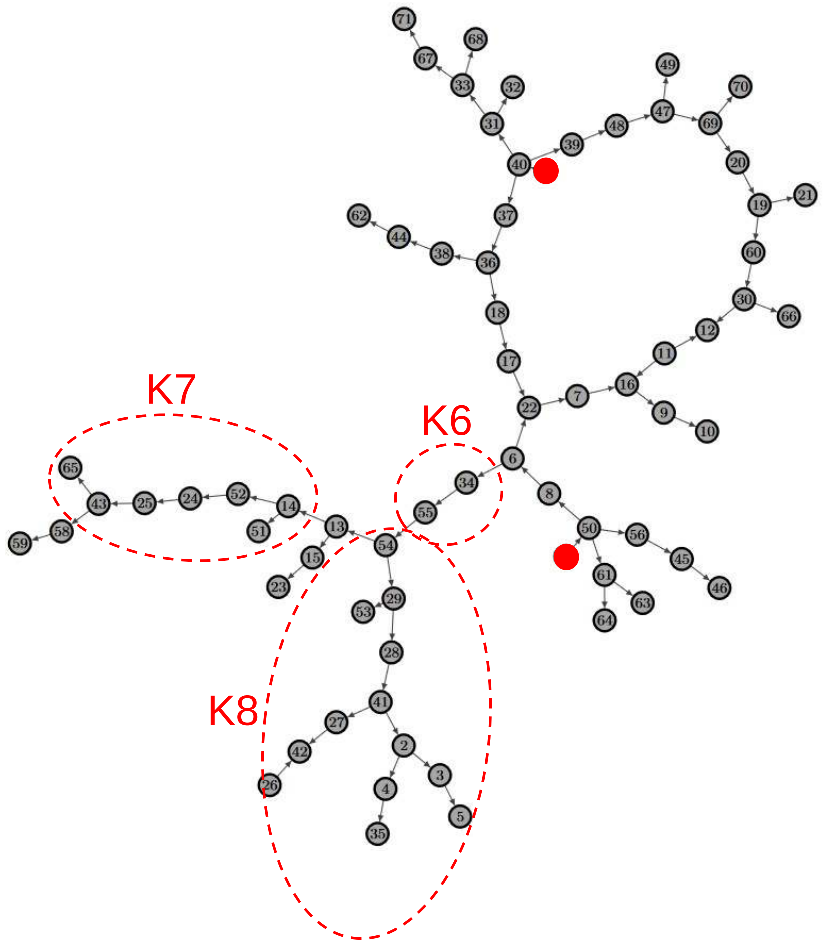


Q3.2 – What are the performances of this new simulation framework?



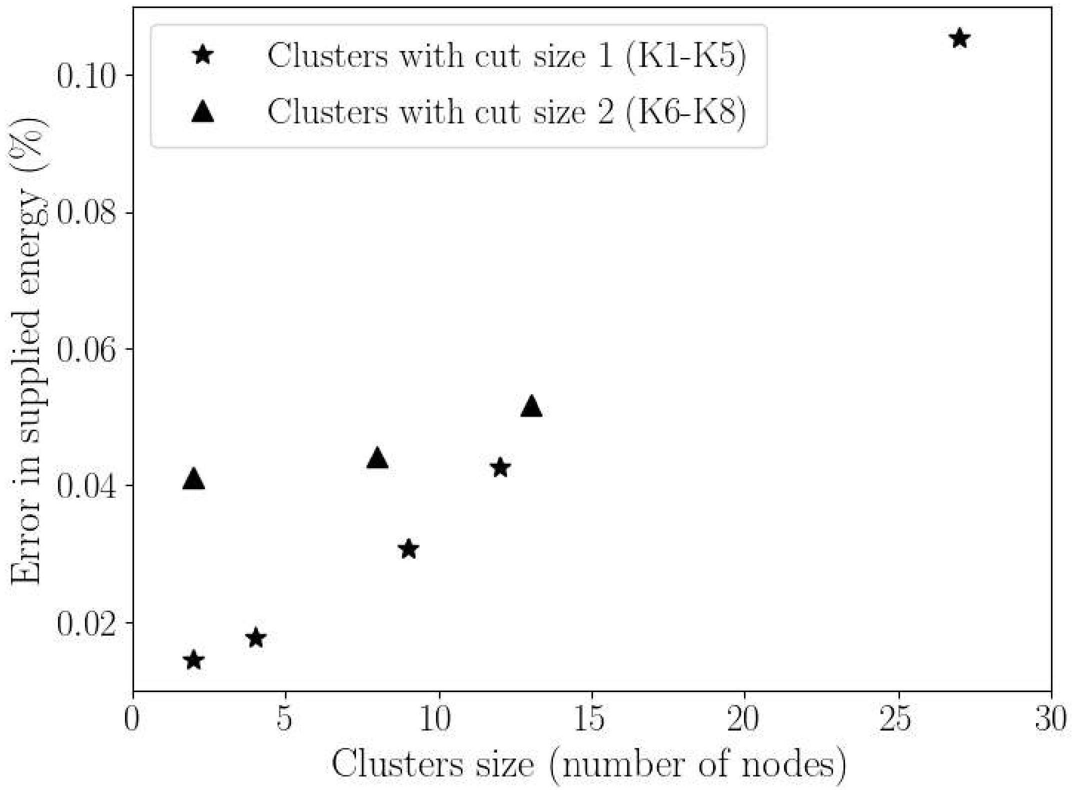
- Baseline: HeatGrid on the **original** DHNs
- 8 different clusters considered:
  - 5 clusters of cut size 1 (K1-K5)
  - 3 clusters of cut size 2 (K6-K8)

Q3.2 – What are the performances of this new simulation framework?

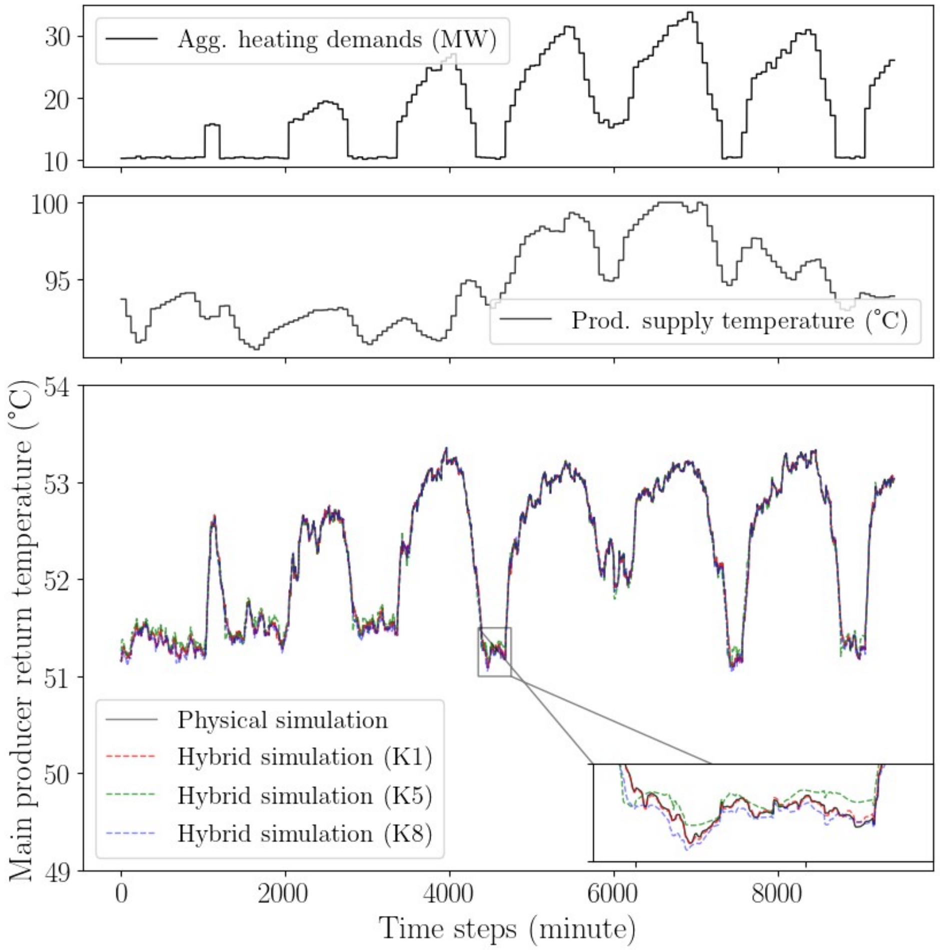


- Baseline: HeatGrid on the **original** DHNs
- 8 different clusters considered:
  - 5 clusters of cut size 1 (K1-K5)
  - 3 clusters of cut size 2 (K6-K8)
- Performances evaluated on:
  - Differences in total energy supplied
  - Differences in total simulation CPU time

Q3.2 – What are the performances of this new simulation framework?



Spearman (size, MAE) = 0.13



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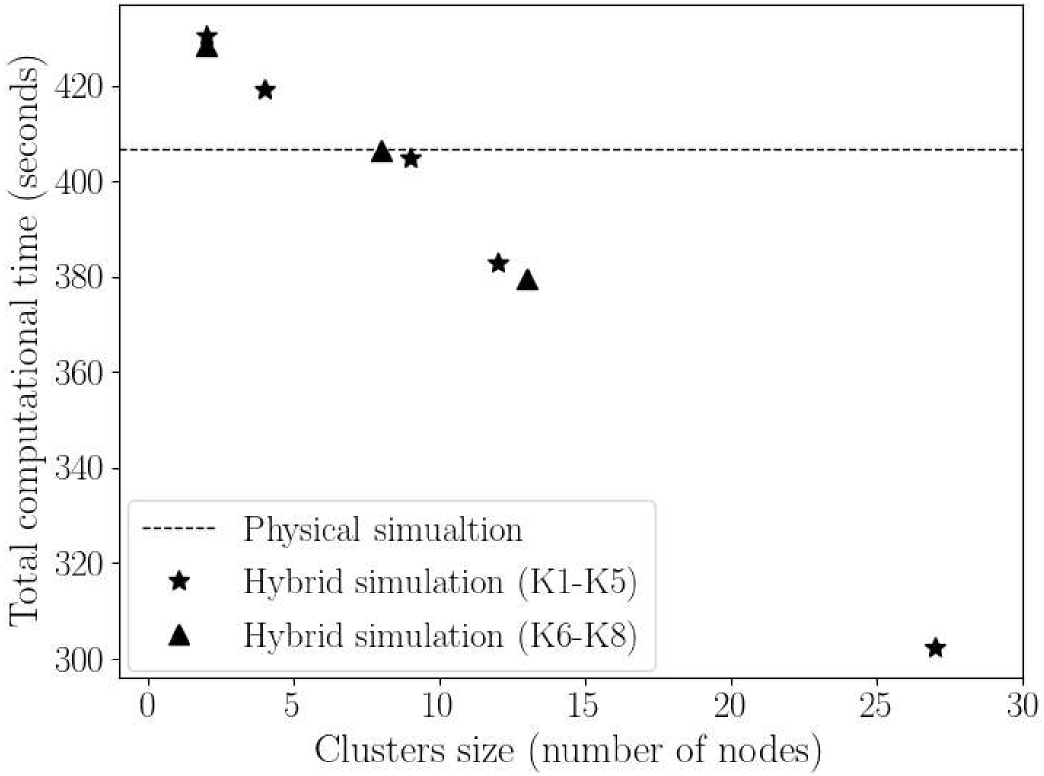
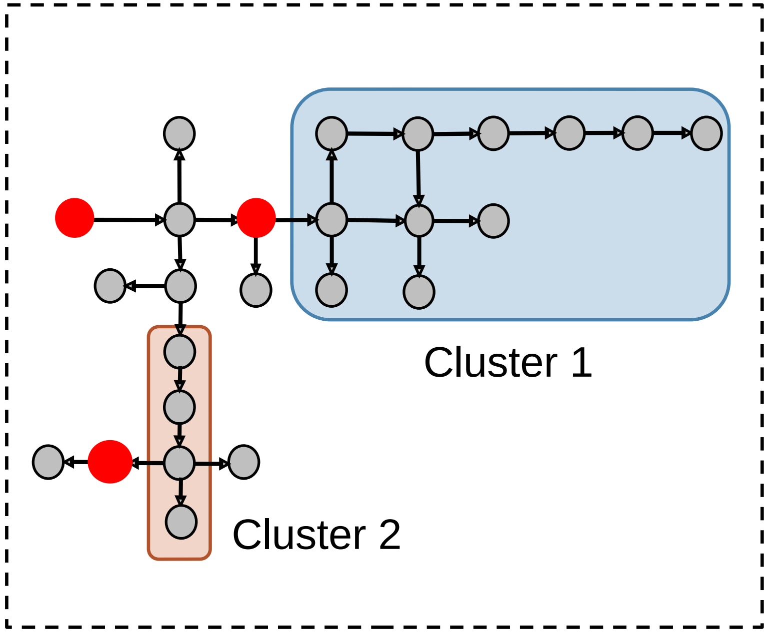


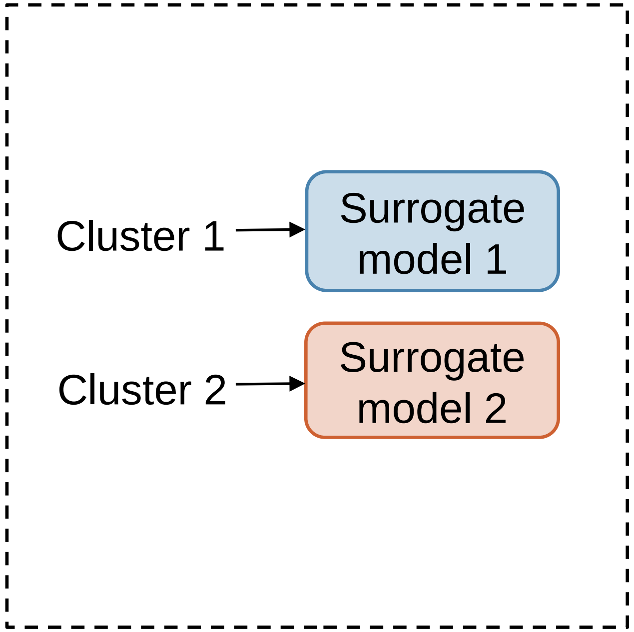
- Replacing **larger** clusters led to more significant **computational reduction**
- Replacing 39% (K5) leads to 27% computational decrease for 0.1% error

Q3.3 – Which strategies allow high accuracy and high computational reduction of this new simulation framework?

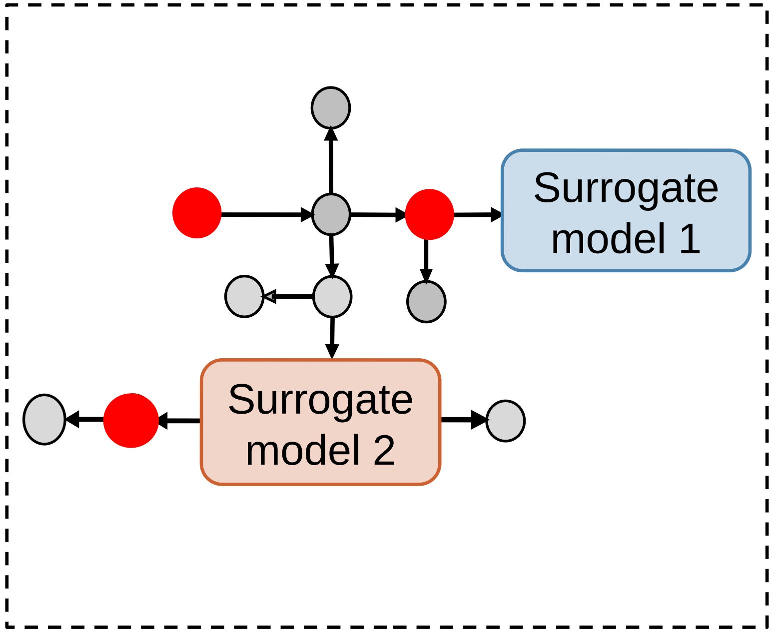
Select clusters of substations



Develop ML surrogates



Simulate reduced network



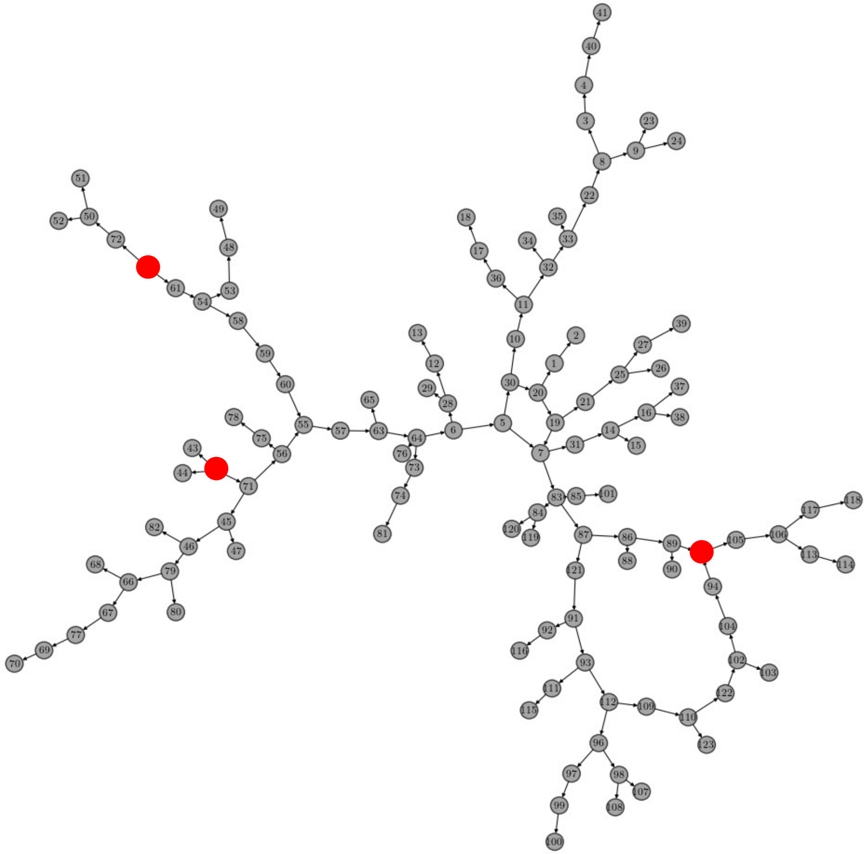
Proposed **clustering**

- Min. Energy Error to Loss
- Max. Spatial reduction

Train **GRU-NN** models

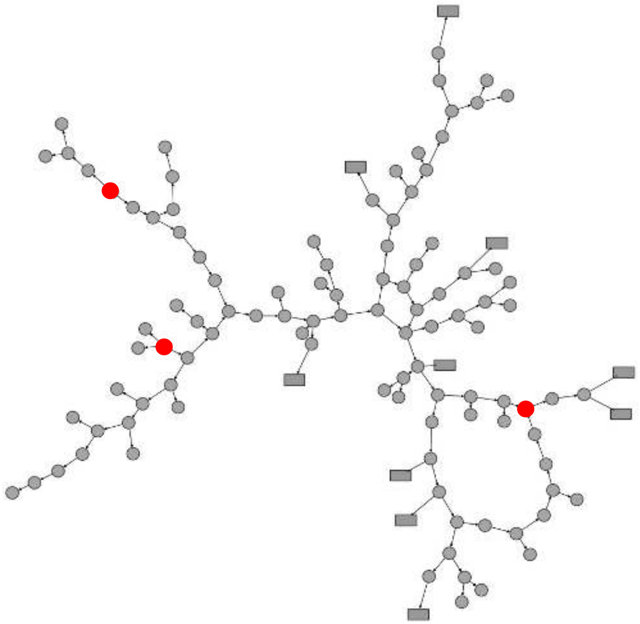
Perform **hybrid simulation**

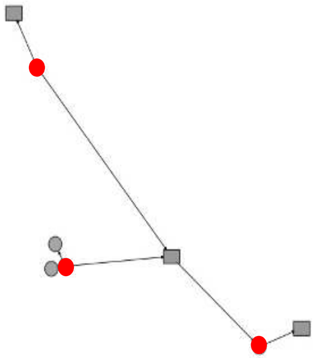
Q3.3 – Which strategies allow high accuracy and high computational reduction of this new simulation framework?



- Unseen validation DHN
- 120 nodes with 3 distributed heat producers

Q3.3 – Which strategies allow high accuracy and high computational reduction of this new simulation framework?





	Min. Energy Error to Loss	**Max. Spatial reduction**
Spatial reduction	16%	96%
Error in energy supplied	**3.76**%	26.37%
Simulation time reduction	7.4% (1.08x)	91% (**11.5**x)

Conclusion

Research Question Q1 - What ML model architectures, features, and training methodologies enable learning and preserving the physical dynamics of substation clusters within DHNs?

Q1.1 ➢ ML surrogate models predict outgoing temperatures from incoming flows information and heating demands

Q1.2 ➢ Surrogate models trained by minimizing MAE between predicted and physical simulated temperatures

Q1.3 ➢ Sequential input GRU-NN architecture balances accuracy and computational efficiency

Research Question Q2 - What properties of substation clusters and unsupervised ML methodologies enable the automatic identification of the optimal clusters that support high accuracy?

Q2.1
- Clusters quality are measured by the surrogate models accuracy (MAE, Energy Error, Energy Error relative to loss)

Q2.2
- High-quality clusters have low external connectivity, minimal transport thermal loss, and short transport delay time

Q2.3
- Distance-based agglomerative clustering identifies automatically best clusters, incorporating user strategy preferences

Research Question Q3 - What strategies enable efficient integration of surrogate ML models into DHNs' dynamic simulations while reducing computational costs and preserving physical accuracy?

Q3.1 ➢ Hybrid Simulation: combining resolution of physical equations with surrogate models predictions

Q3.2 ➢ Hybrid Simulation accuracy assessed by low errors in conserving the total energy supplied, with computational gains reflected by reduction in total simulation CPU time

Q3.3 ➢ User strategy preferences can be included within the clustering in order to favor accurate or rapid simulation.

Limits and Perspectives

- **Trusts** in the ML surrogate models predictions
  - Features Importance using Explainable AI (Saliency maps*, LRP, LIME)
  - Quantify predictions uncertainty (Bayesian Neural Network)

- **Impacts of position** of the clusters within the network
  - Quantifying the relative position of clusters (Shortest path)

- **Cluster-specific** ML surrogate models
  - Graph Foundation Models (*e.g.,* Grid FM)

*preliminary results

Thank You

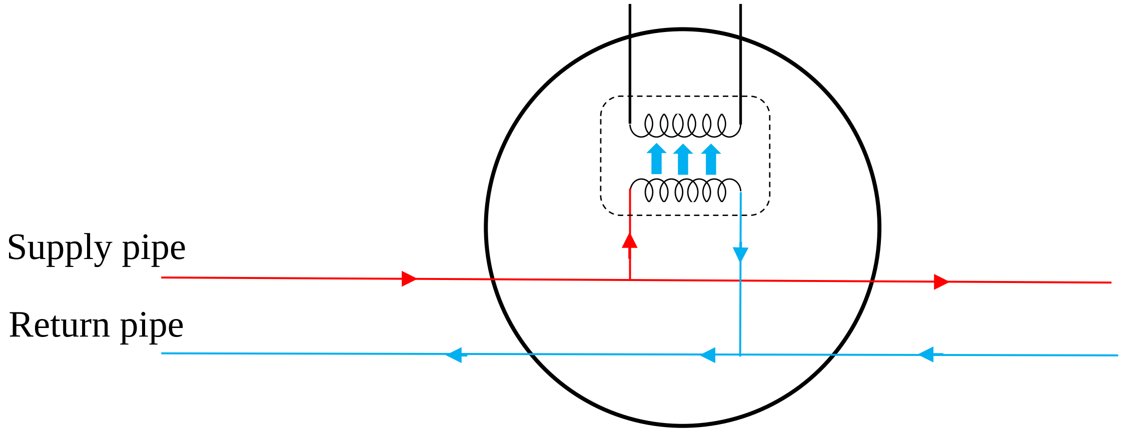
For your Attention

3 Journal articles (1 published, 1 in press, 1 in submission)
2 Conference articles
1 Workshop article

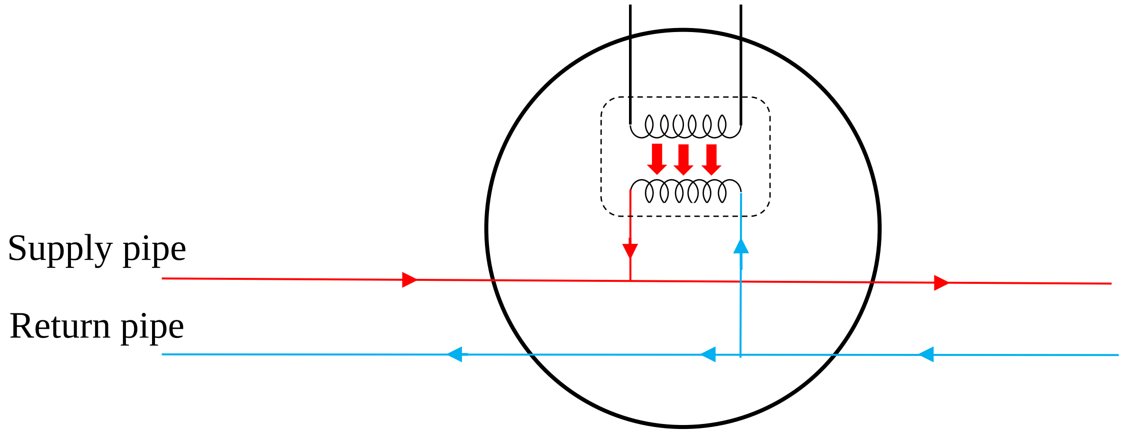
1 Open-source software (“SyntheticDHN”)
Open-access repositories (github: drod-96)

Appendices

Modeling substations and producers



Modeling of a **substation** with a heating demand D



Modeling of a **heat producer** with a production power P_S

Synthetic heating demand profile generation

Heating law based generation

$$D_i^t = \max(U_i A_i (T_{thershold} - T_{ext}^t), 0.2\text{Peak}_i)$$

A_i: exchange area (random choice)

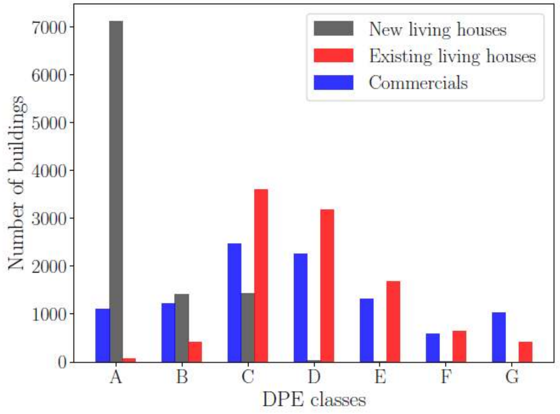
U_i: U-factor value

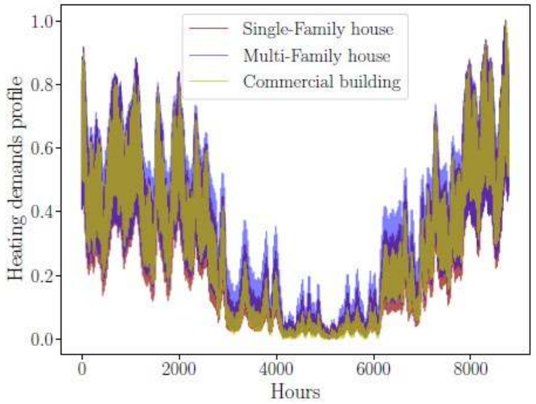
Ancient Buildings: 3.4 $\mathrm{W.m^{-2}.K^{-1}}$

Recent Buildings: 0.84 $\mathrm{W.m^{-2}.K^{-1}}$

Offices Buildings: 2.5 $\mathrm{W.m^{-2}.K^{-1}}$

DPE based approach





When2Heat Heating Profiles

- Reflecting France's buildings performances distribution*
- Use class energy ranges (consumption per year)

$$E_i = \int_{t=1}^{8760} D_i^t \, dt$$

$$D_i^t = \gamma v^t$$

*ADEME data for 2019

Annexes

DHN simulation models

HeatGrid

$$\forall i \in V, \sum_j \dot{m}^{j,i}_{in_s} + m^i_s = \sum_l \dot{m}^{i,l}_{out_s} + m^i_c$$

$$H^j - H^i = k_{i,j} \left|\dot{m}^{i,j}_{in_s}\right| \dot{m}^{i,j}_{in_s}$$

$$\frac{\partial T}{\partial t} = -\frac{4\dot{m}}{\pi\rho d^2}\frac{\partial T}{\partial x} - \frac{4\boldsymbol{h}}{\rho c_p d}(T - T_{ext})$$

Tin → Tout

PyDHN (Boghetti & Kämpf, Energy, 2024)

$$\sum_{j=1}^{m} b_{ij}\Delta p_j = 0 \quad \forall i = 1, \dots l, \quad b_{ij} \in \boldsymbol{B}$$

$$\boldsymbol{F}(\boldsymbol{X}) = \sum_{j=1}^{m} b_{ij}\phi_j(b^T_{ji}\dot{m}_{c,i}) = 0 \quad \forall i = l^* + 1, \dots l$$

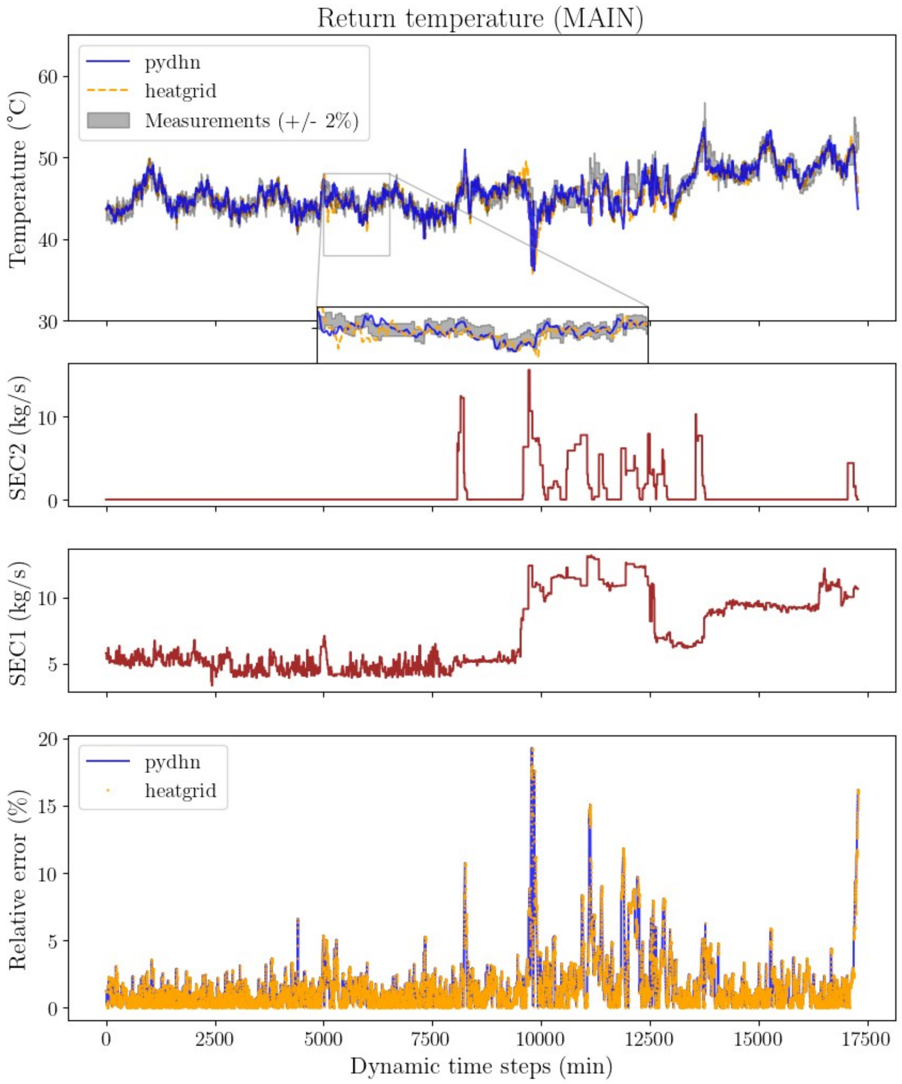
$$\begin{bmatrix} \frac{dT_{wc}}{dt} \\ \frac{dT_B}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{h_B}{\rho_w A_{wc} c_{p,w}} & \frac{h_B}{\rho_w A_{wc} c_{p,w}} \\ \frac{h_B}{C_B} & -\frac{h_B + h_{ins}}{C_B} \end{bmatrix} \begin{bmatrix} T_{wc} \\ T_B \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{h_{ins}}{C_B} T_{ext} \end{bmatrix}$$

$$\begin{aligned} T_{wc}(x,0) &= \theta_{wc}(x,t) \\ T_B(x,0) &= \theta_B(x,t) \end{aligned}, \tag{2.19}$$

where $h_B = 1/\left(\frac{1}{h_w} + \frac{1}{h_s}\right)$ and $C_B = \rho_w A_{wb} c_{p,w} + \rho_s A_s c_{p,s}$.

Tin → Tout →

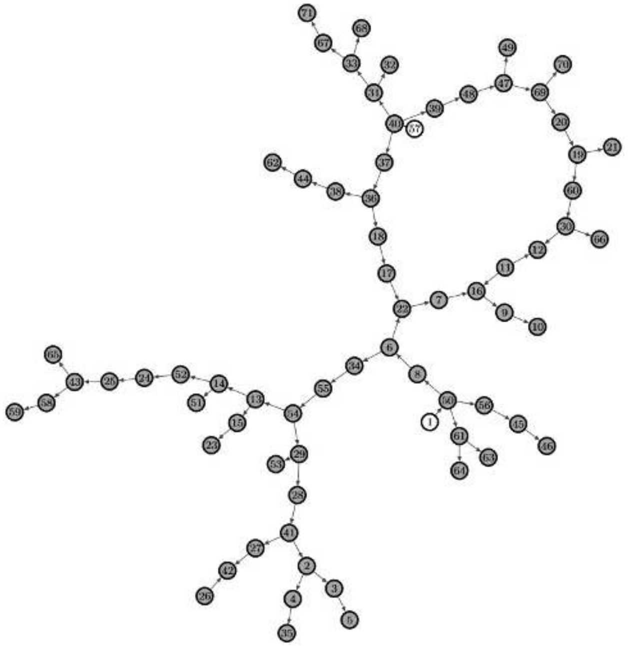
DHN simulation models (results)

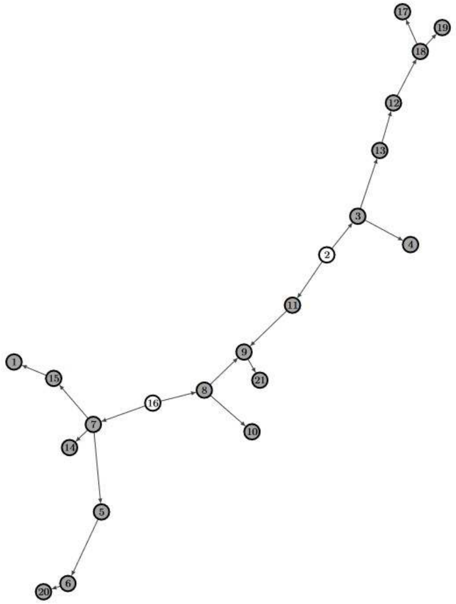


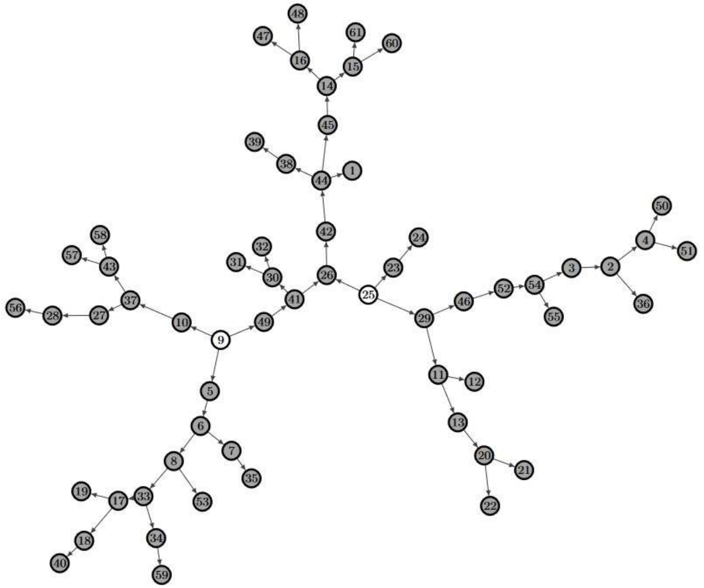
Comparison	MAE (°C)	MARE (%)
PyDHN vs Measurements		
HeatGrid vs Measurements		

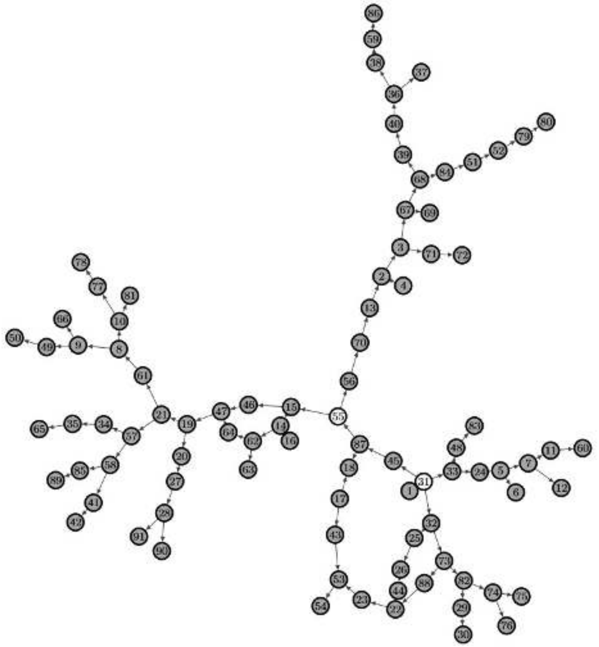
Comparison	MAE (°C)	MARE (%)
PyDHN vs Measurements	0.73 °C	1.57 %
HeatGrid vs Measurements	0.89 °C	1.93 %

Case study DHNs for Research Question 1







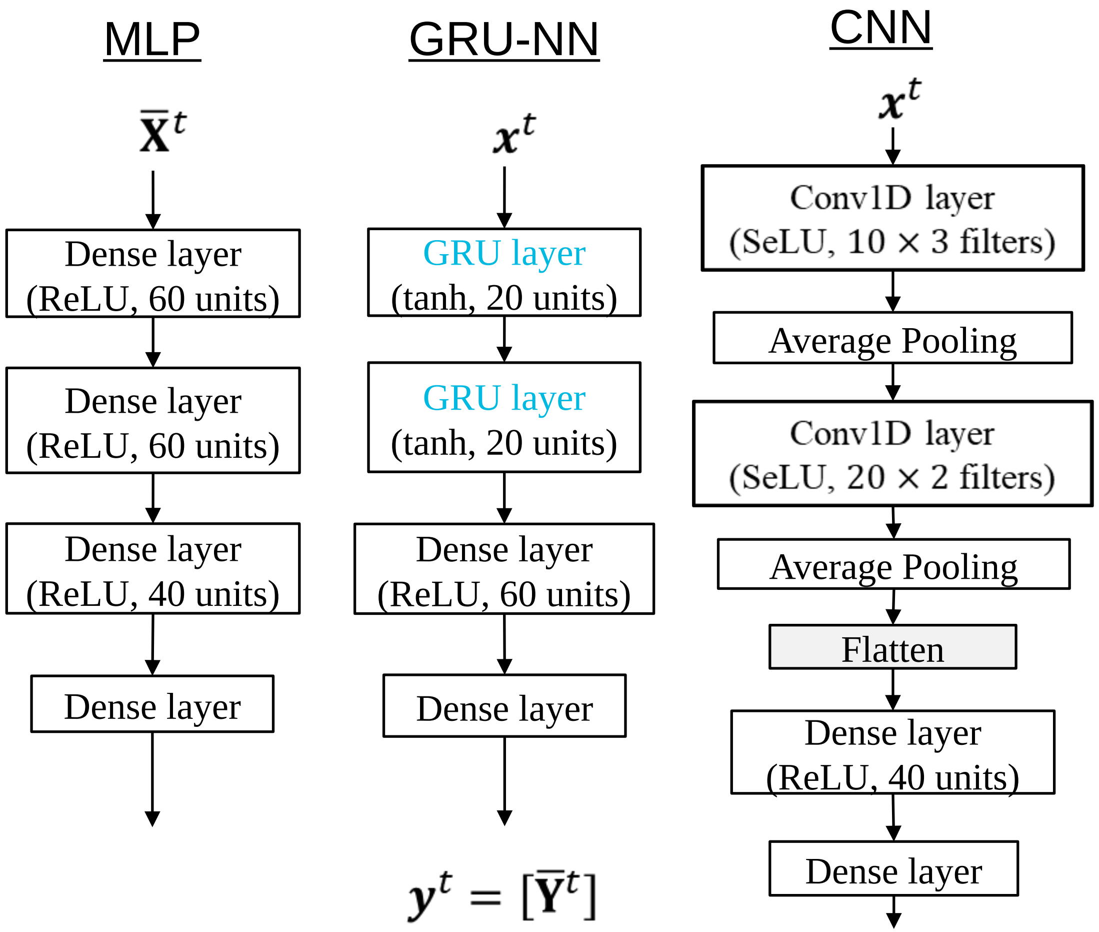


- Four different case study networks considered
- 220 substation clusters manually selected

ML Architectures Compared

$$\boldsymbol{x}^t = \begin{cases} [\overline{\mathbf{X}}^{t-N_{steps/seq}-1}, \ldots, \overline{\mathbf{X}}^t] & \text{for GRU-NN and CNN} \\ [\overline{\mathbf{X}}^t] & \text{for MLP} \end{cases}$$

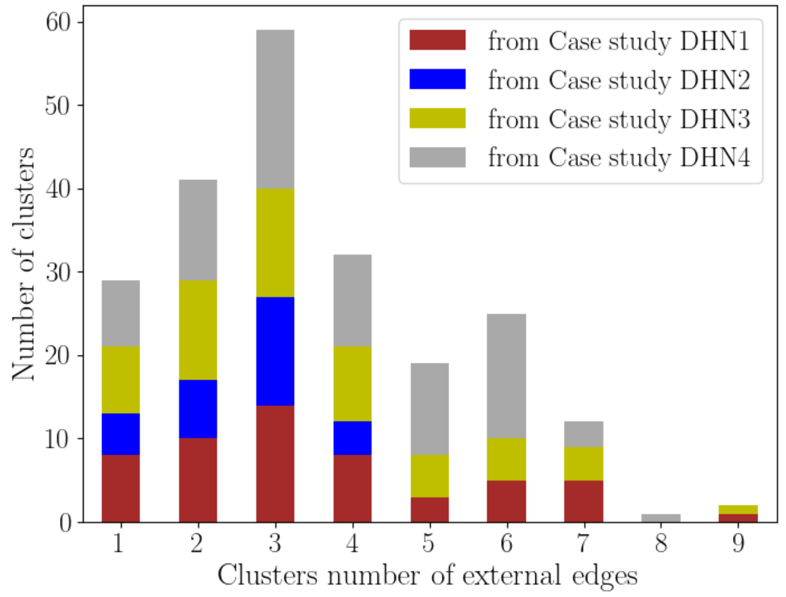
$N_{steps/seq}$: Input context length

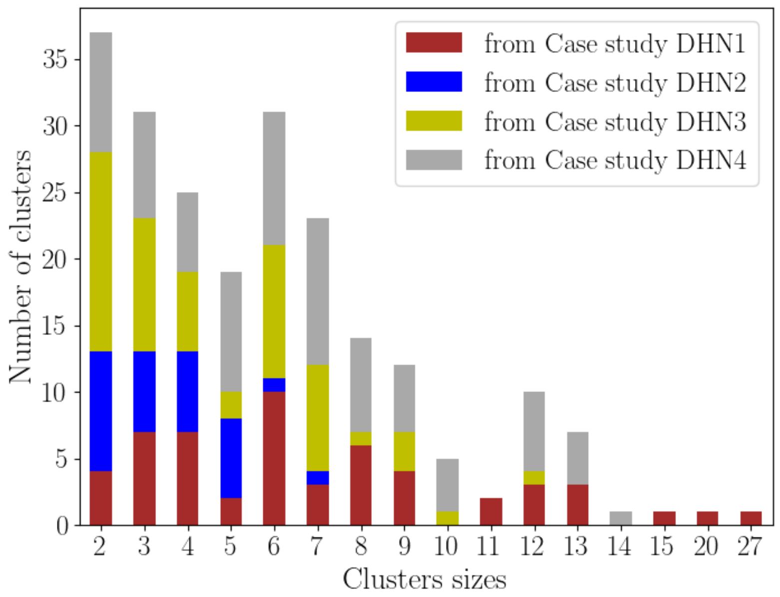


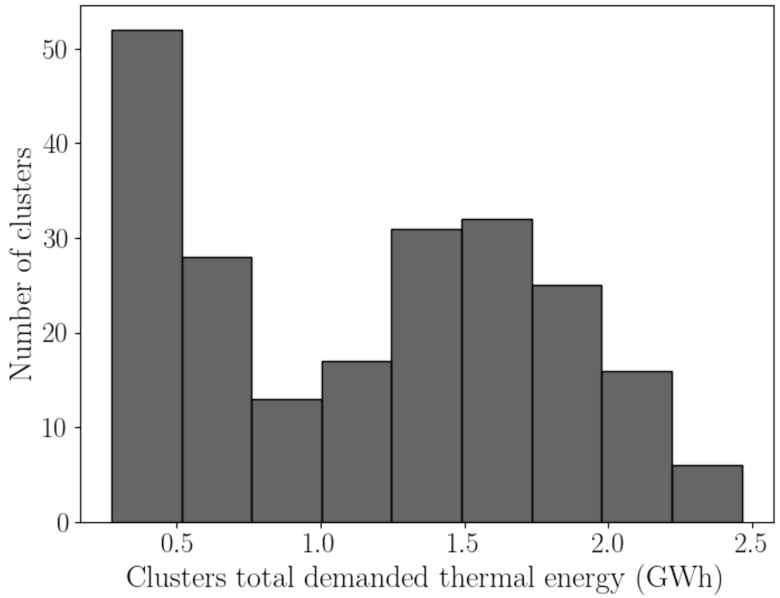
Training Procedures Details

- Trained for 100 epochs, learning rate starting at 10^{-3} with step change of 10^{-1}/20 epochs with ADAM optimizer
- No dropout with 10^{-6} weight decay
- Features are normalized using Min-Max
- Hyperparameters tuning using Hyperband techniques
- Hyperparameters tuning using Hyperband techniques

Case Study Selected Clusters



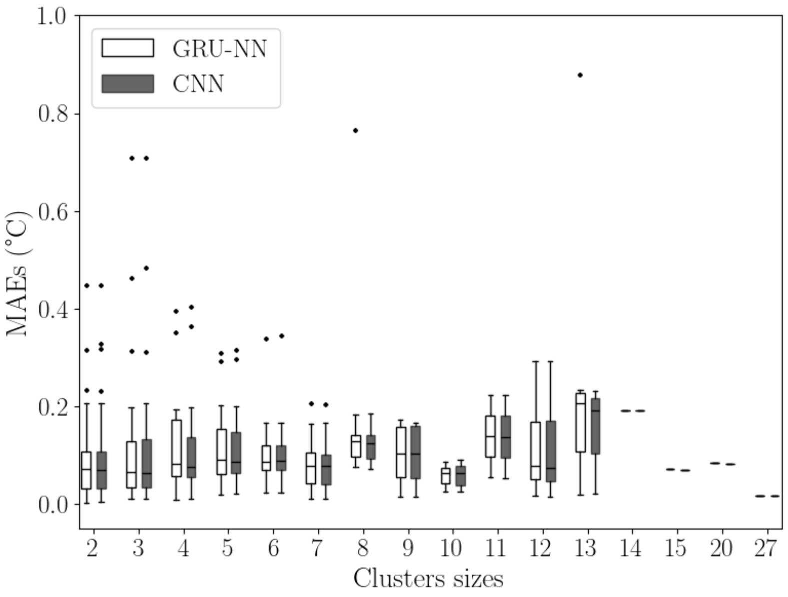


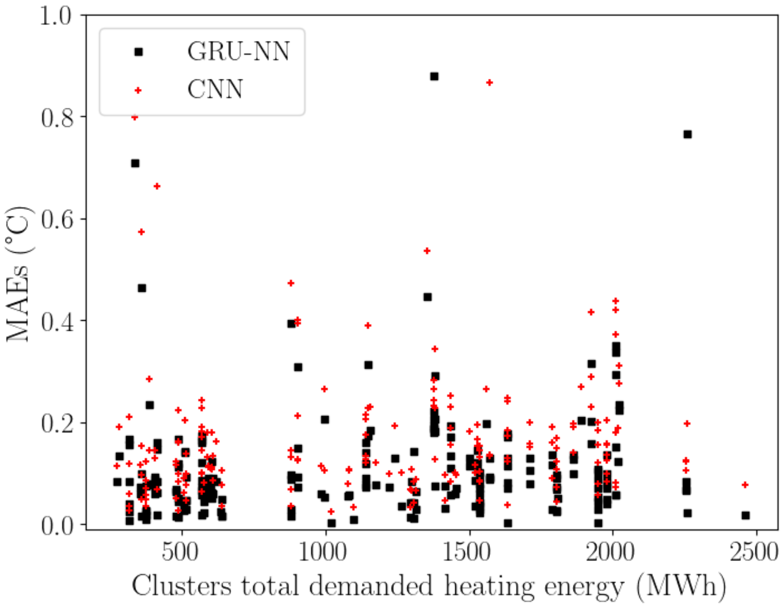


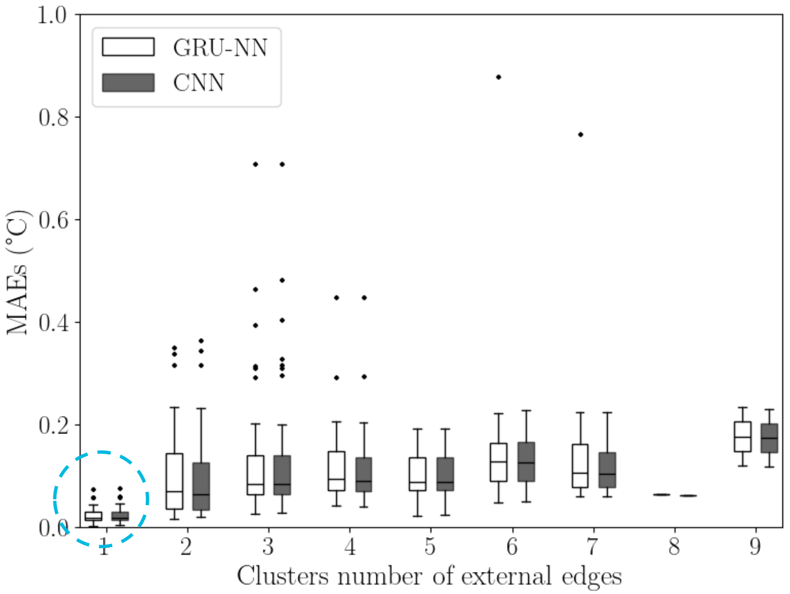
- Selected clusters display large variability in topology (size* and external edges)
- Aggregated heating demand levels of the clusters occupy also wide range of values between 27MWh to 2500 MWh.

*Size: number of composing nodes

Distribution of Performances





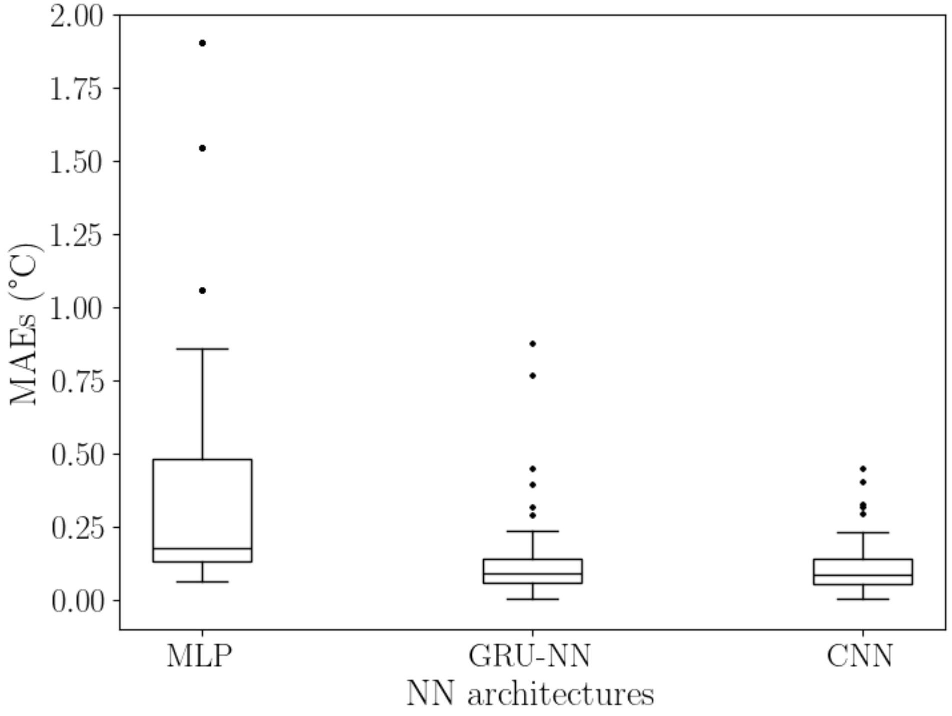


Consistent performances across clusters with difference sizes and heating consumption levels

Drop of performances for **number of external edges >** 1

Annexes

ML Architectures Performances – Statistical tests

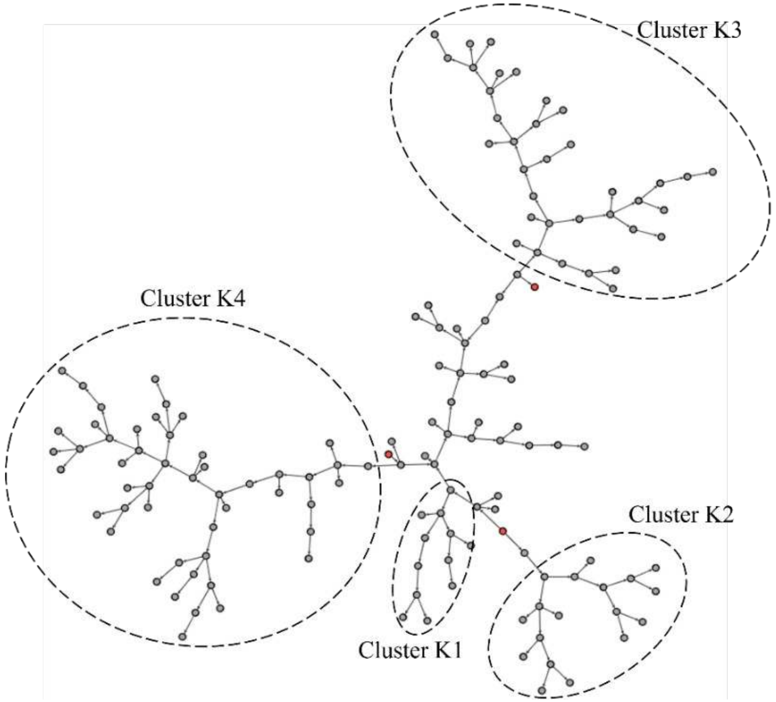


MAEs distribution comparison test	p-value
MLP > GRU-NN	7.10^{-27}
MLP > CNN	3.10^{-23}
GRU-NN > CNN	0.695

- Distributions differences validated using non-parametric Mann-Whitney U-test[Fay+10] statistical test
- Null hypothesis: no statical differences between the distribution with p-value $\leq$ 0.05 validating alternative

Model Predictive Control (MPC) – Result 1

$$\min_{\mathbf{M}_s, \mathbf{T}_{ss}} \sum_{t=1}^{N_t} \left[\sum_{\{i \in S\}} \dot{\mathbf{M}}_{s_i}^{t} c_p \left(\mathbf{T}_{ss_i}^{t} - T_{sr_i} \right) \right] + 0 \left(\max\left(0, \left(\delta_{surplus} - 2\right)\right)^2 + \max\left(0, \left(\delta_{deficits} - 2\right)\right)^2 \right) \qquad \mathbf{T}_{ss} \in [80,90]°C$$

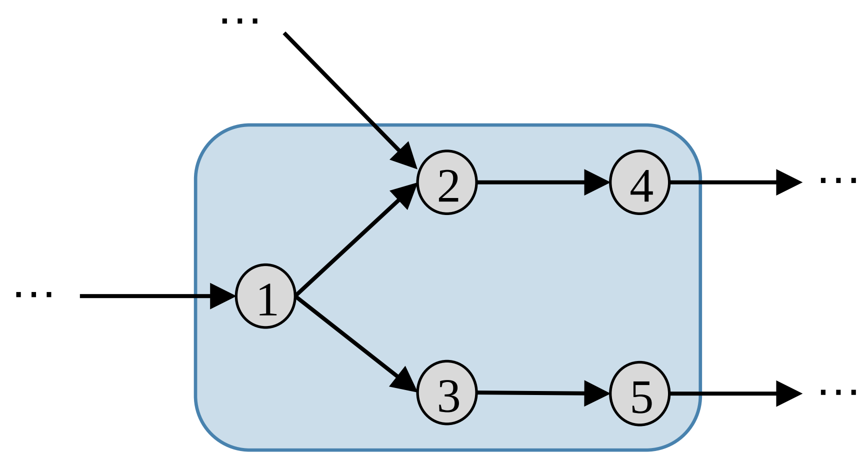


	Physical MPC	Hybrid MPC				Rule-based
Clusters	-	K1	K2	K3	K4	-
Clustered (%)	0	8.6	12	24	33	0
Produced energy (MWh)						
Total CPU time (s)	8593	8590	8210	8001	7650	-

	Physical MPC	Hybrid MPC				Rule-based
Clusters	-	K1	K2	K3	K4	-
Clustered (%)	0	8.6	12	24	33	0
Produced energy (MWh)	202.8	206.6	202.7	203.2	203.6	235.7
$\delta_{deficits}$	1.46	1.37	1.44	1.38	1.40	1.32
$\delta_{surplus}$	0.41	0.68	0.39	0.40	0.44	5.94
Total CPU time (s)	8593	8590	8210	8001	7650	-

- GRU-NN used with additional output feature (real consumption of the clusters)
- Decisions every 1h with 3h sliding window

How to compute the Mean/Max transport delay time?



Cluster $\mathcal{C}$

Delay time through an edge

$$\tau_{(i,j)} = \frac{\rho\left(\pi d_{(i,j)}^2 l_{(i,j)}\right)}{4\dot{m}_{(i,j)}}$$

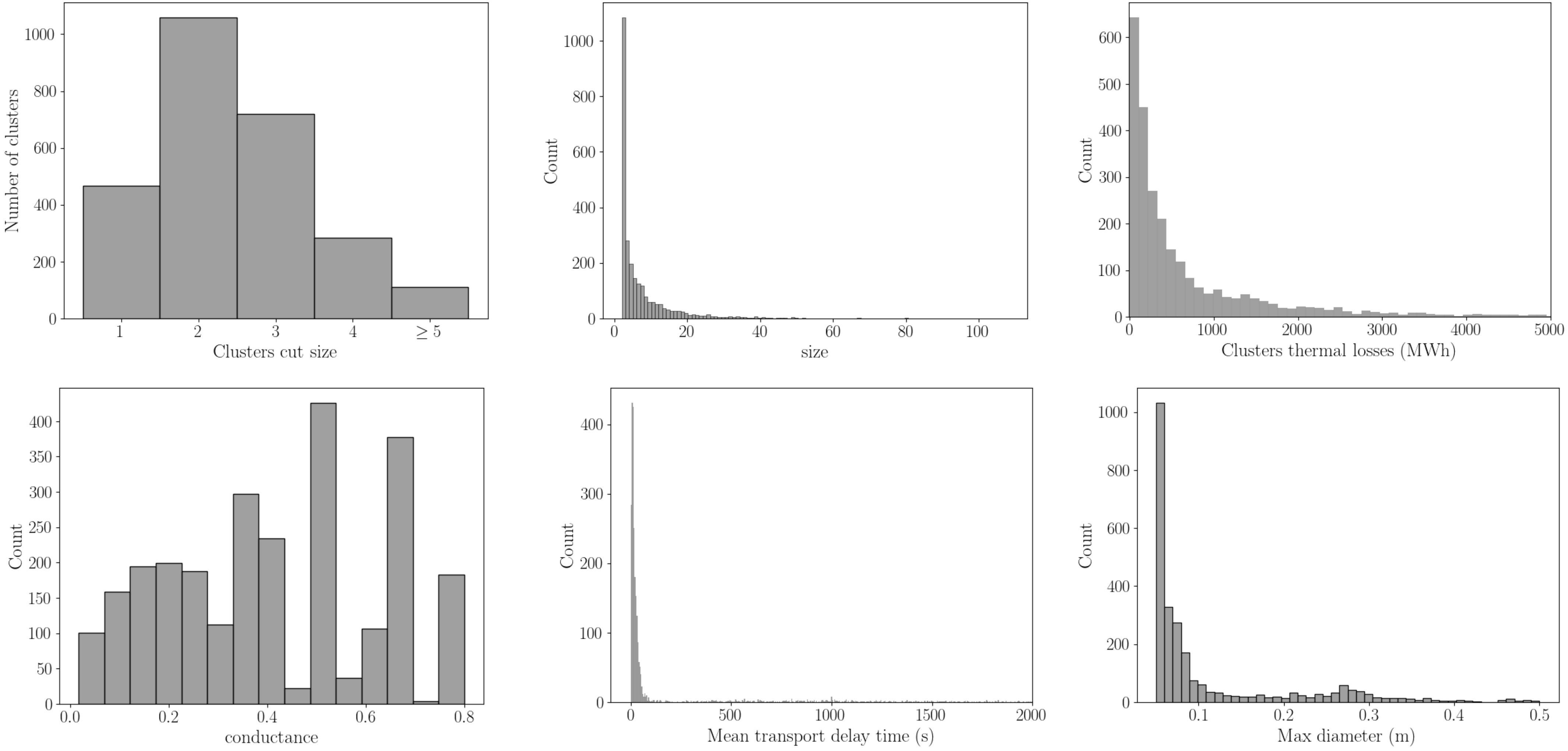
Compute cluster all transport delay times $\boldsymbol{\tau_C}$

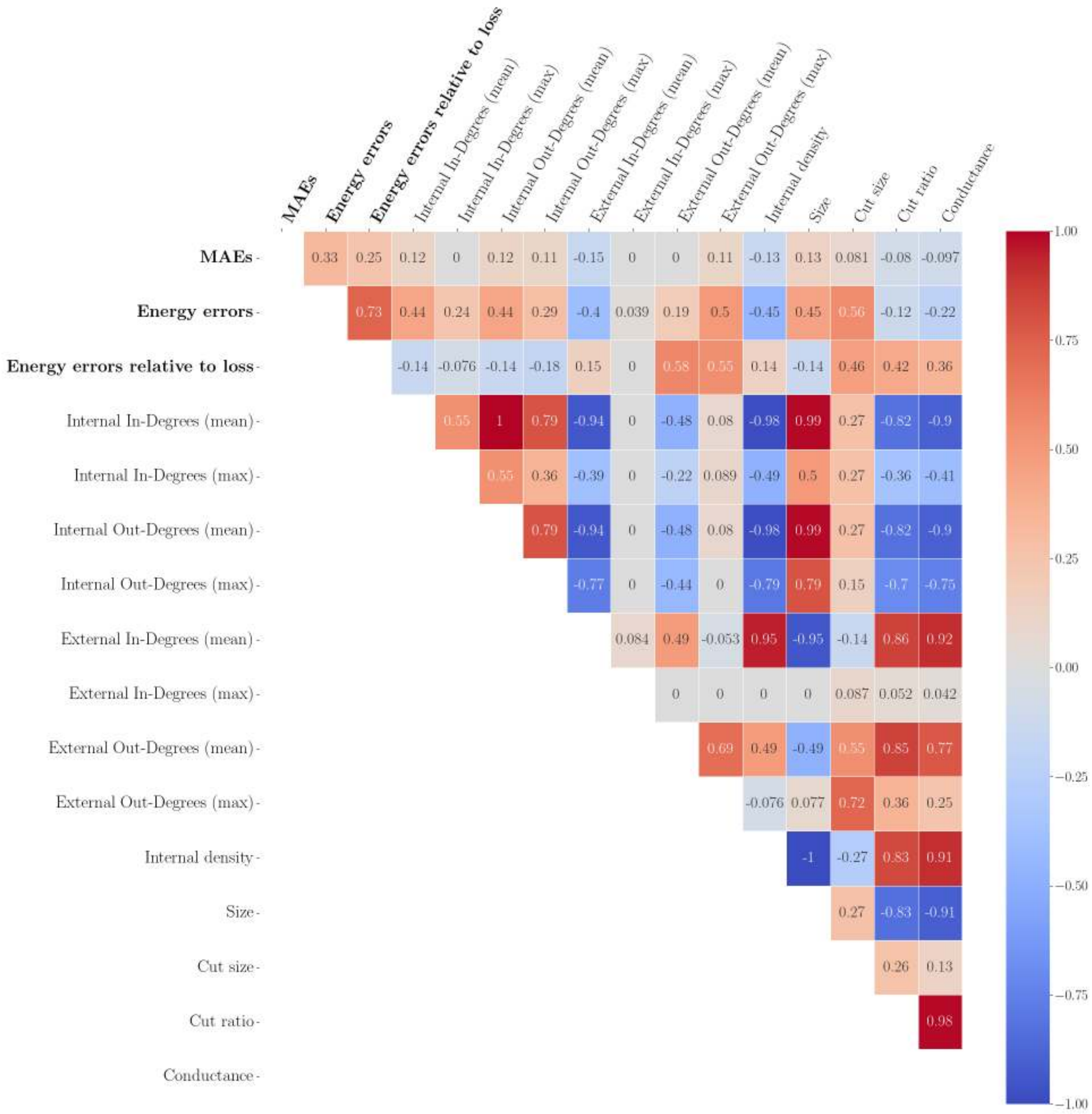
- Assign edges weights to $\tau_{(i,j)}$
- Djikstra algorithm between all pairs of interface nodes

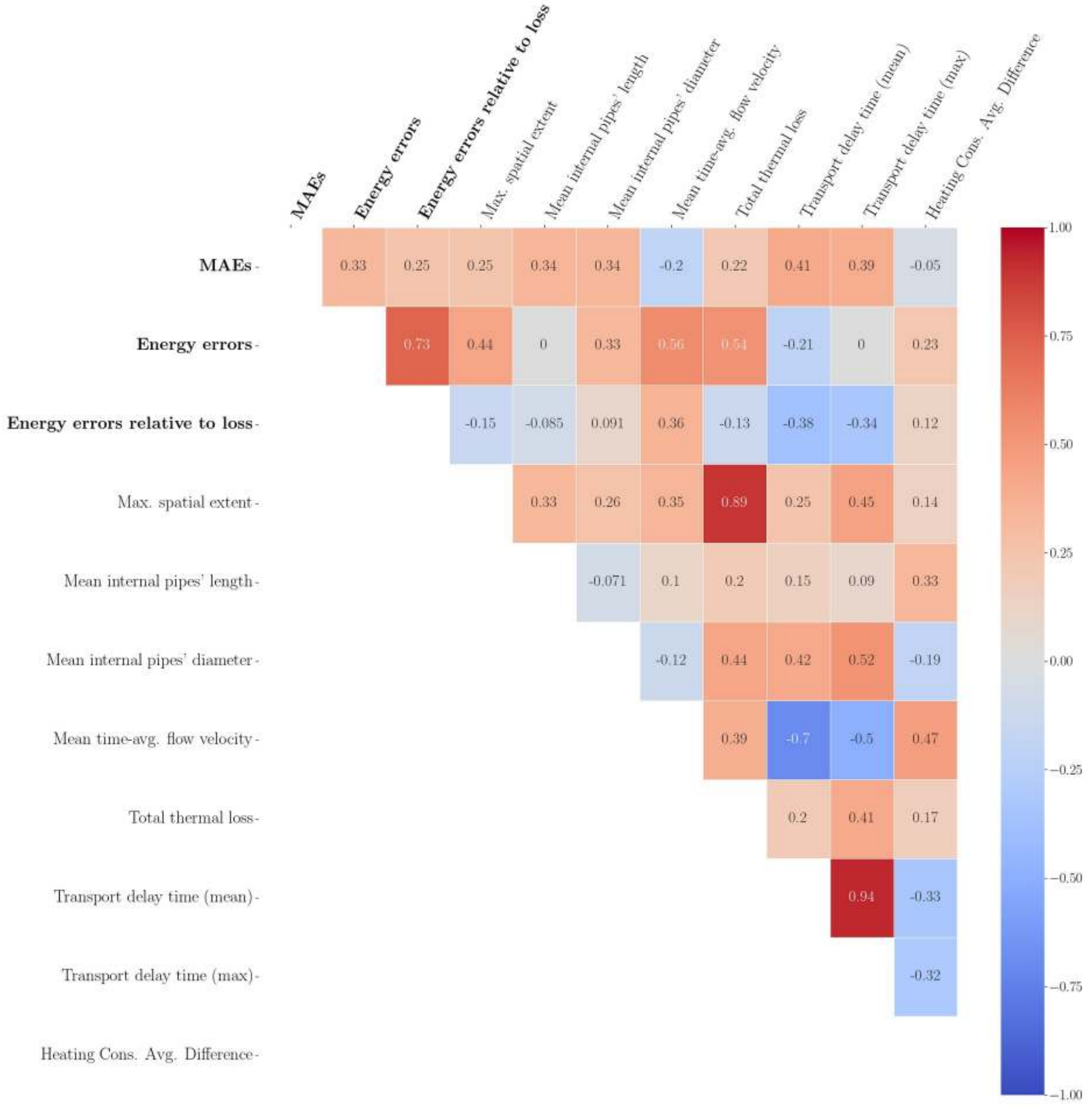
Compute mean/max values of $\boldsymbol{\tau_C} = \{\boldsymbol{\tau_{1\to5}}, \boldsymbol{\tau_{1\to4}}, \cancel{\boldsymbol{\tau_{2\to4}}}, \boldsymbol{\tau_{2\to5}}\}$

Annexes

Selected Clusters Properties distribution

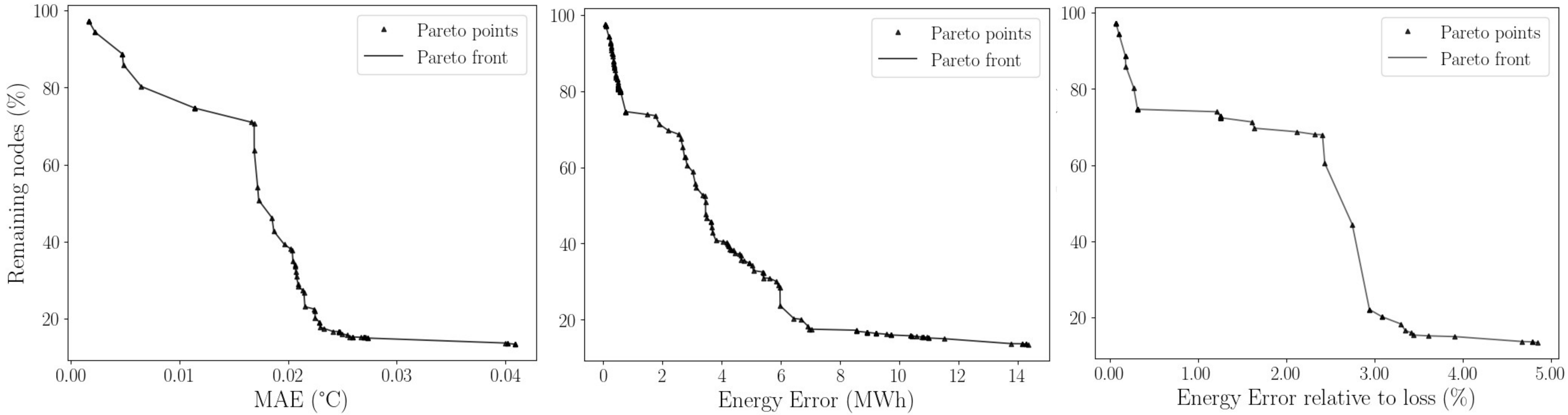


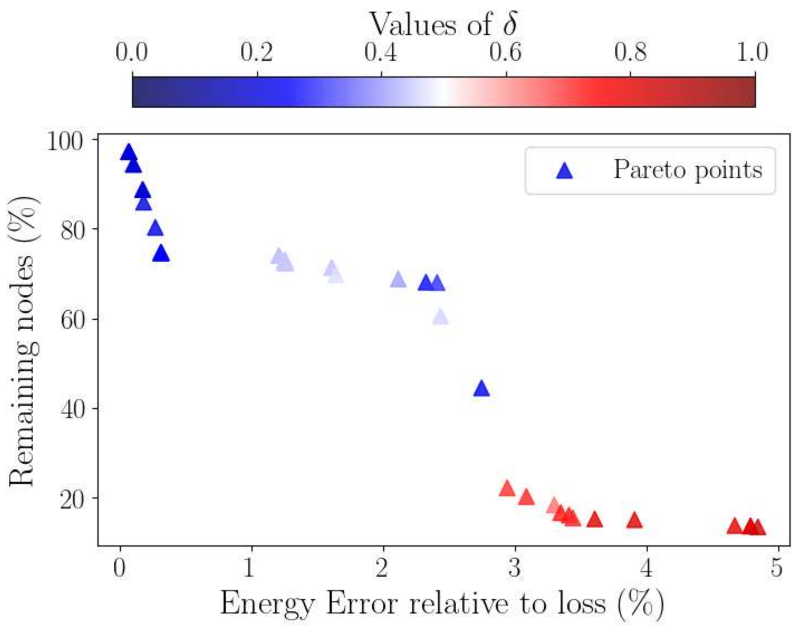

MAEs
Energy errors
Energy errors relative to loss
Internal In-Degrees (mean)
Internal In-Degrees (max)
Internal Out-Degrees (mean)
Internal Out-Degrees (max)
External In-Degrees (mean)
External In-Degrees (max)
External Out-Degrees (mean)
External Out-Degrees (max)
Internal density
Size
Cut size
Cut ratio
Conductance
MAEs: 0.33 0.25 0.12 0 0.12 0.11 -0.15 0 0 0.11 -0.13 0.13 0.081 -0.08 -0.097
Energy errors: 0.73 0.44 0.24 0.44 0.29 -0.4 0.039 0.19 0.5 -0.45 0.45 0.56 -0.12 -0.22
Energy errors relative to loss: -0.14 -0.076 -0.14 -0.18 0.15 0 0.58 0.55 0.14 -0.14 0.46 0.42 0.36
Internal In-Degrees (mean): 0.55 1 0.79 -0.94 0 -0.48 0.08 -0.98 0.99 0.27 -0.82 -0.9
Internal In-Degrees (max): 0.55 0.36 -0.39 0 -0.22 0.089 -0.49 0.5 0.27 -0.36 -0.41
Internal Out-Degrees (mean): 0.79 -0.94 0 -0.48 0.08 -0.98 0.99 0.27 -0.82 -0.9
Internal Out-Degrees (max): -0.77 0 -0.44 0 -0.79 0.79 0.15 -0.7 -0.75
External In-Degrees (mean): 0.084 0.49 -0.053 0.95 -0.95 -0.14 0.86 0.92
External In-Degrees (max): 0 0 0 0 0.087 0.052 0.042
External Out-Degrees (mean): 0.69 0.49 -0.49 0.55 0.85 0.77
External Out-Degrees (max): -0.076 0.077 0.72 0.36 0.25
Internal density: -1 -0.27 0.83 0.91
Size: 0.27 -0.83 -0.91
Cut size: 0.26 0.13
Cut ratio: 0.98
1.00 0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75 −1.00

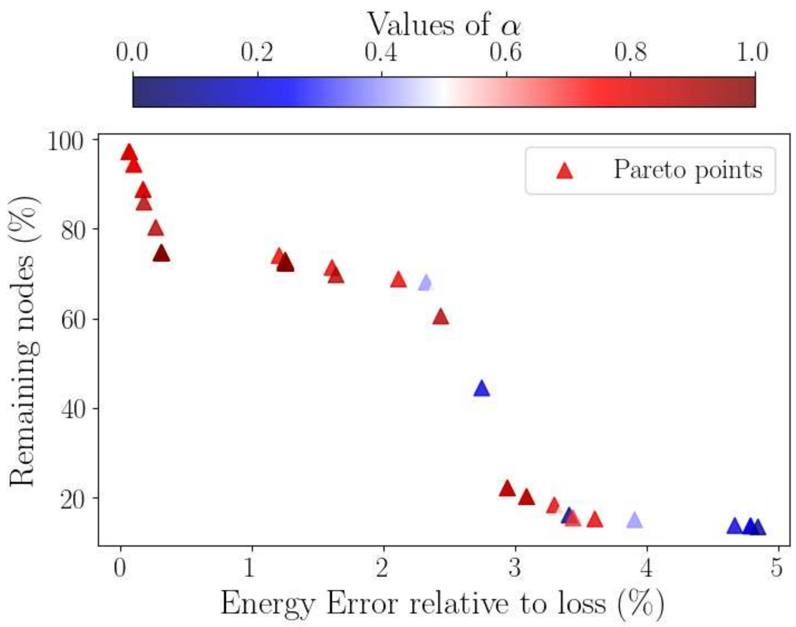

MAEs
Energy errors
Energy errors relative to loss
Max. spatial extent
Mean internal pipes' length
Mean internal pipes' diameter
Mean time-avg. flow velocity
Total thermal loss
Transport delay time (mean)
Transport delay time (max)
Heating Cons. Avg. Difference
MAEs: 0.33 0.25 0.25 0.34 0.34 -0.2 0.22 0.41 0.39 -0.05
Energy errors: 0.73 0.44 0 0.33 0.56 0.54 -0.21 0 0.23
Energy errors relative to loss: -0.15 -0.085 0.091 0.36 -0.13 -0.38 -0.34 0.12
Max. spatial extent: 0.33 0.26 0.35 0.89 0.25 0.45 0.14
Mean internal pipes' length: -0.071 0.1 0.2 0.15 0.09 0.33
Mean internal pipes' diameter: -0.12 0.44 0.42 0.52 -0.19
Mean time-avg. flow velocity: 0.39 -0.7 -0.5 0.47
Total thermal loss: 0.2 0.41 0.17
Transport delay time (mean): 0.94 -0.33
Transport delay time (max): -0.32
1.00 0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75 −1.00

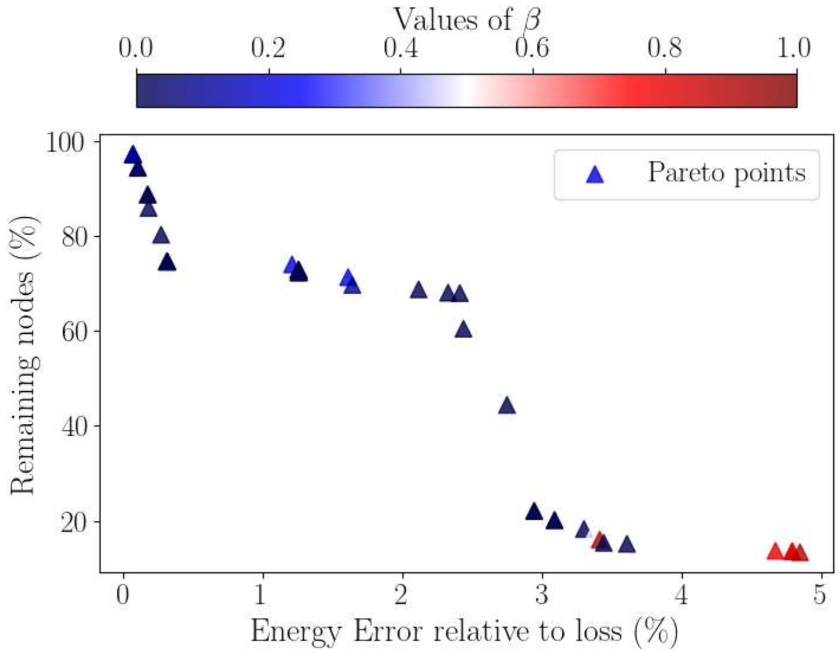
Annexes

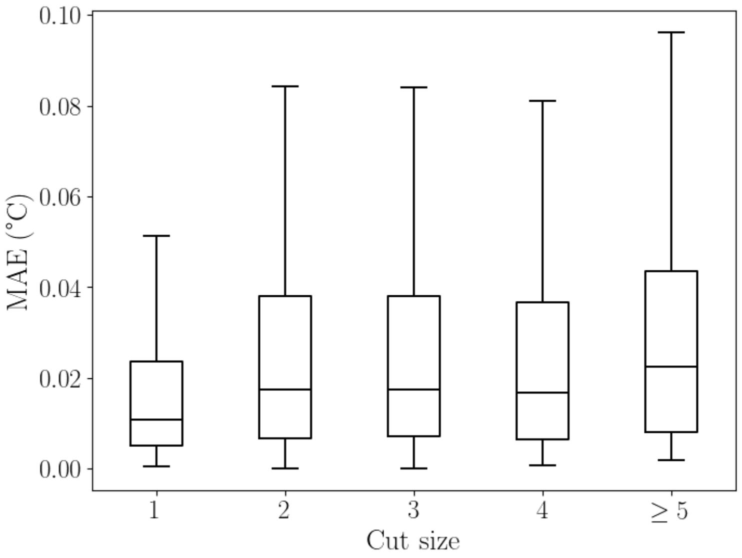
Pareto Fonts

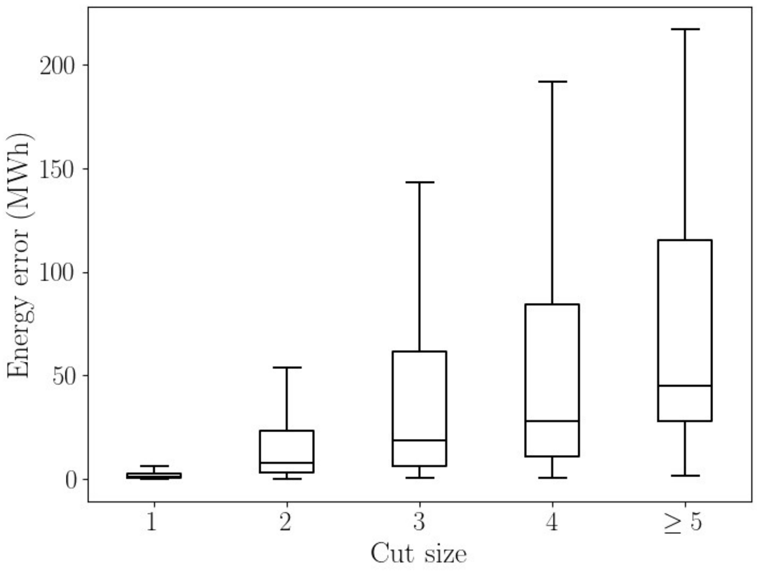


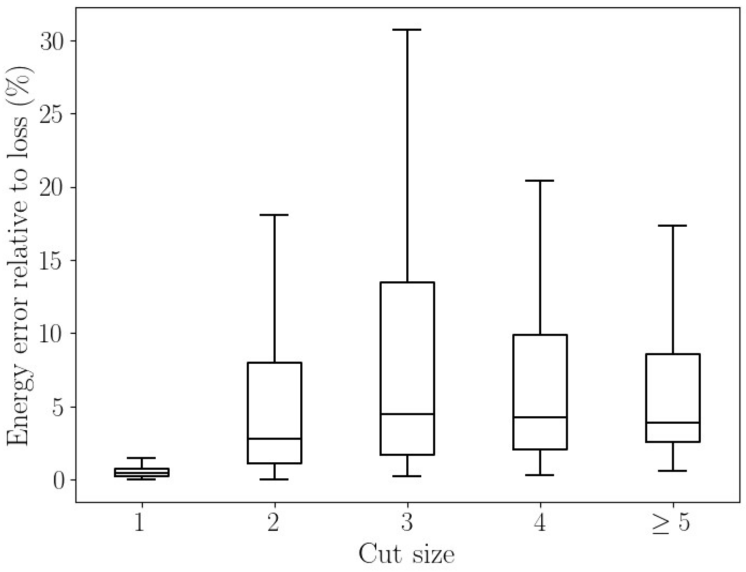

Values of δ
0.0 0.2 0.4 0.6 0.8 1.0
Pareto points
Remaining nodes (%)
Energy Error relative to loss (%)


Values of α
0.0 0.2 0.4 0.6 0.8 1.0
Pareto points
Remaining nodes (%)
Energy Error relative to loss (%)


Values of β
0.0 0.2 0.4 0.6 0.8 1.0
Pareto points
Remaining nodes (%)
Energy Error relative to loss (%)

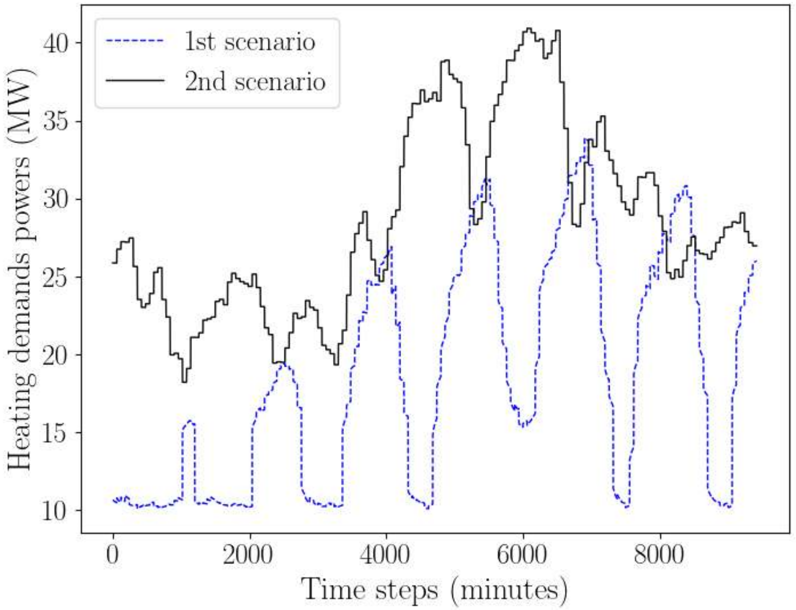


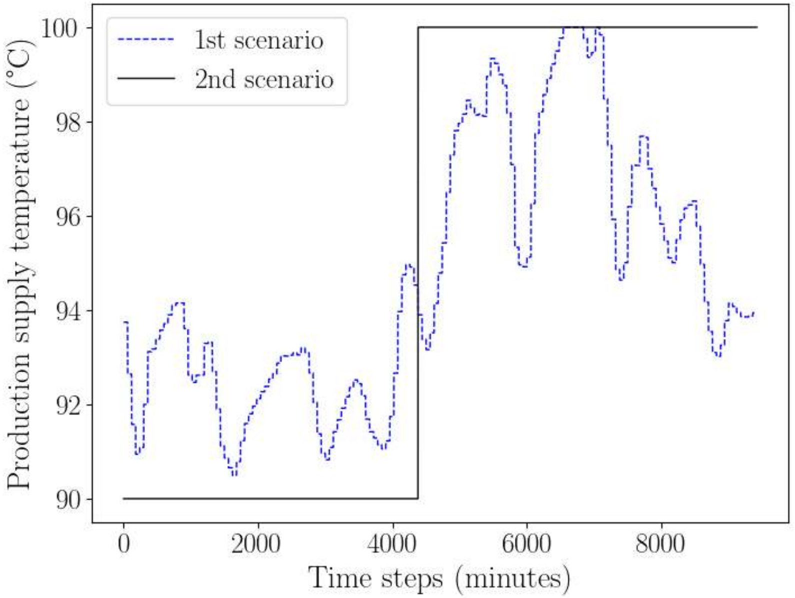




Cut size effects on quality metrics

2 Operating scenarios (case study 1)

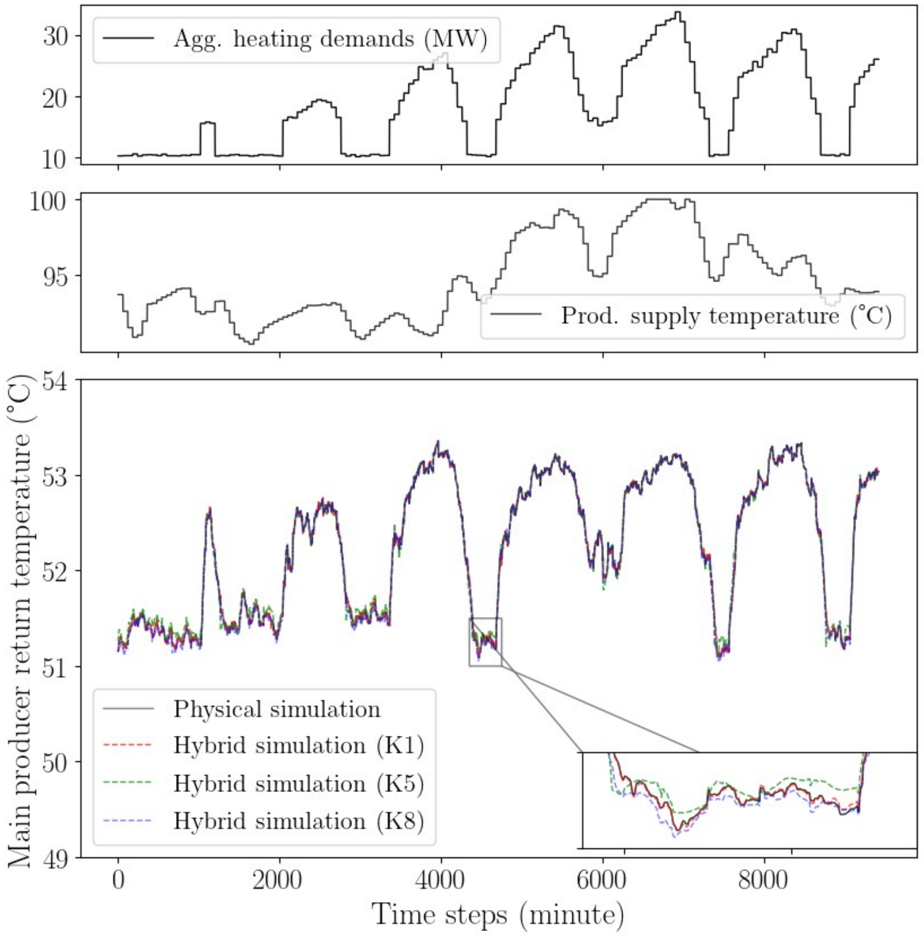


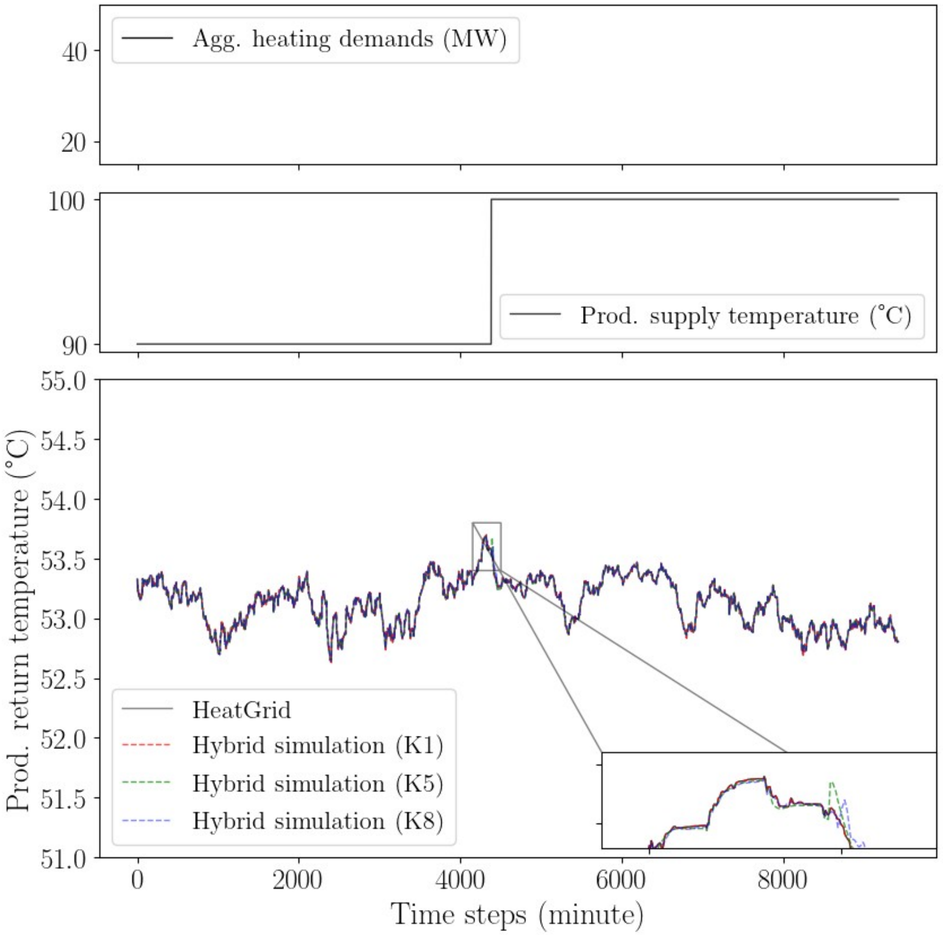


	Scenario 1	Scenario 2
*Change in production temperature profile	No (in August*)	Yes (in March*)

*surrogate models trained on heating law production temperature
[90-100]°C, with heating data between January and June 2019

2 Operating scenarios (case study 1)



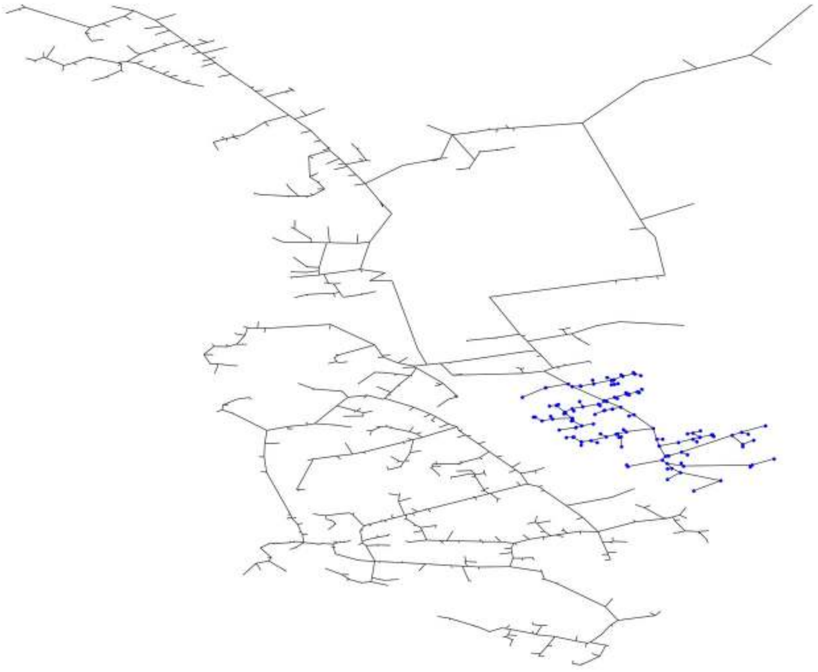


Hybrid Simulation Overall Results (case study 1)

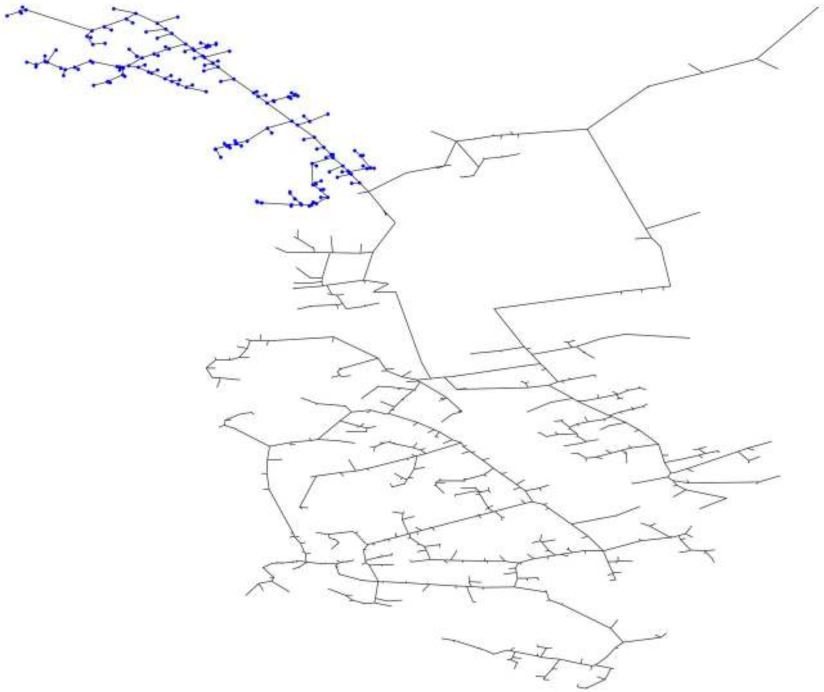
Cluster replaced	**K1**	**K2**	**K3**	**K4**	**K5**	**K6**	**K7**	**K8**
Sizes	2	4	9	12	27	2	8	13
Cut sizes	1	1	1	1	1	2	2	2
Surrogate Models MAE (°C)	0.032	0.010	0.017	0.017	0.017	0.016	0.075	0.019
Return temp. MAE (in 0.1°C) 1st scenario	0.05	0.06	0.10	0.15	0.37	0.12	0.14	0.17
Return temp. MAE (in 0.1°C) 2nd scenario	0.008	0.008	0.03	0.04	0.10	0.08	0.08	0.09
Prod. total energy error (in 0.1%) 1st scenario	0.15	0.18	0.31	0.43	1.06	0.41	0.44	0.52
Prod. total energy error (in 0.1%) 2nd scenario	0.02	0.02	0.08	0.13	0.25	0.25	0.22	0.22

Case study 2

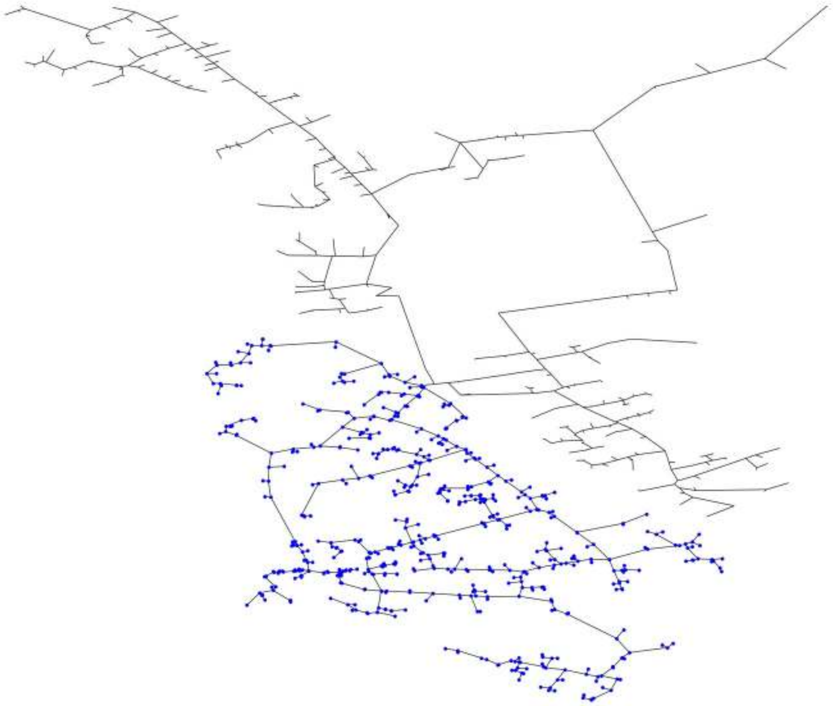
- Large-scale DHN with 1110 nodes (1 main producers)
- **Larger** clusters with different sizes (K1, K2 and K3), replaced individually and jointly



Cluster K1 (130 nodes)



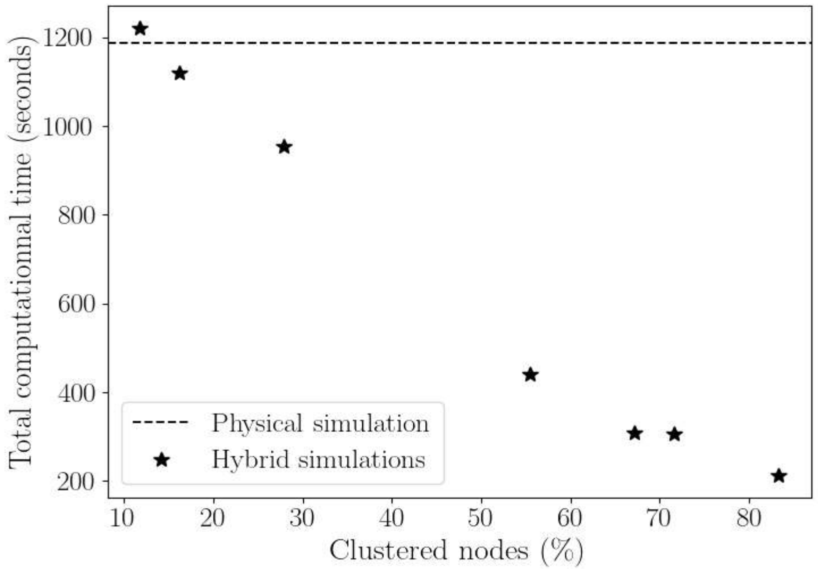
Cluster K2 (180 nodes)



Cluster K3 (615 nodes)

Hybrid Simulation (case study 2) – Results

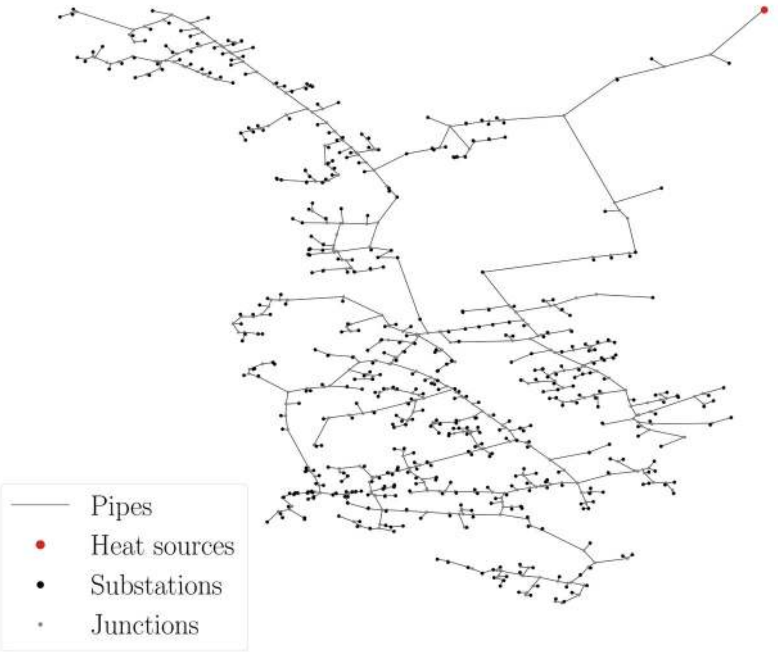
Replaced cluster	**K1**	**K2**	**K1+K2**	**K3**	**K1+K3**	**K2+K3**	**K1+K2+K3**
Sizes	130	180	310	615	745	795	925
Clustered nodes (%)	11.7	16.2	27.9	55.4	67.1	71.6	83.3
Return temp. MAE (in °C)	0.04	0.06	0.21	0.20	0.57	0.21	0.46
Prod. Energy error (in % of loss)	1.24	1.66	6.43	5.97	17.57	6.41	13.23



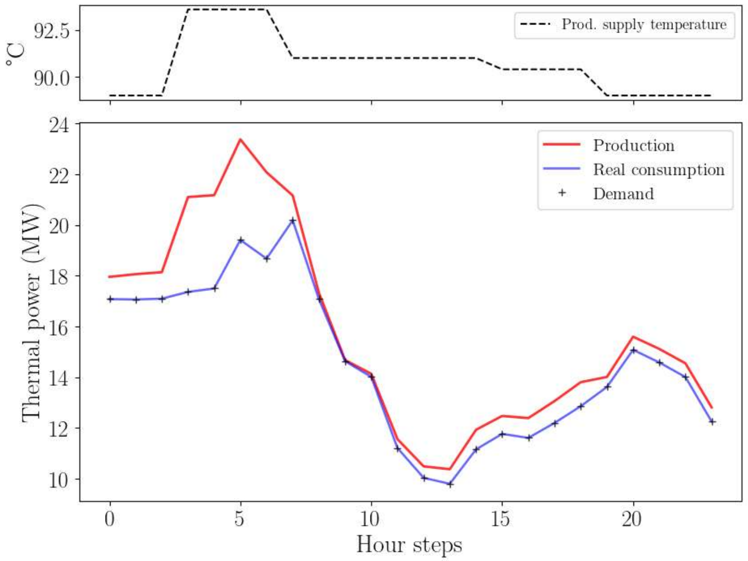
Model Predictive Control (MPC) – Result 2

Minimize $\mathcal{L} = \int_{t_0}^{t_f} (P_{loss}^t)dt + O_1 \int_{t_0}^{t_f} \left(\sum_i^N \delta_i^{+t} \right) dt + O_2 \int_{t_0}^{t_f} \left(\sum_i^N \delta_i^{-t} \right) dt$

$T_{ss} \in [85 - 95]$ °C



Optimal solution found

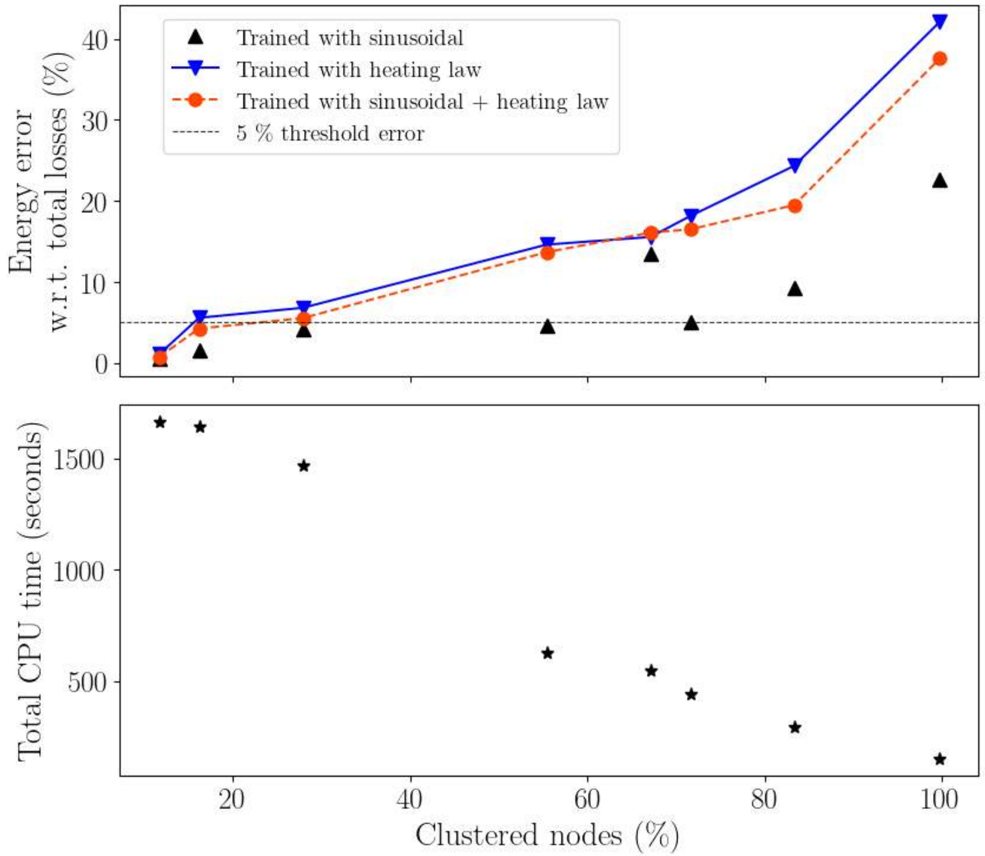


- Optimization for 48 h (eval 24h)
- CMAE-ES ($\mu = 3, \lambda = 6$, 170 iterations)
- **Hybrid-based opt. CPU time: ~ 1.5 days**
- Full physical-based opt. CPU time: > 4 days

→ 3X faster than using physical simulation

K2+K3 replaced: best tradeoff between speed and accuracy

Effects of training data temperature profiles



- Training the models with **sinusoidal, heating law or both** production temperature profiles
- Evaluate the models on sinusoidal and heating law production temperature profiles
- Models trained with sinusoidal profile have better generalizability.

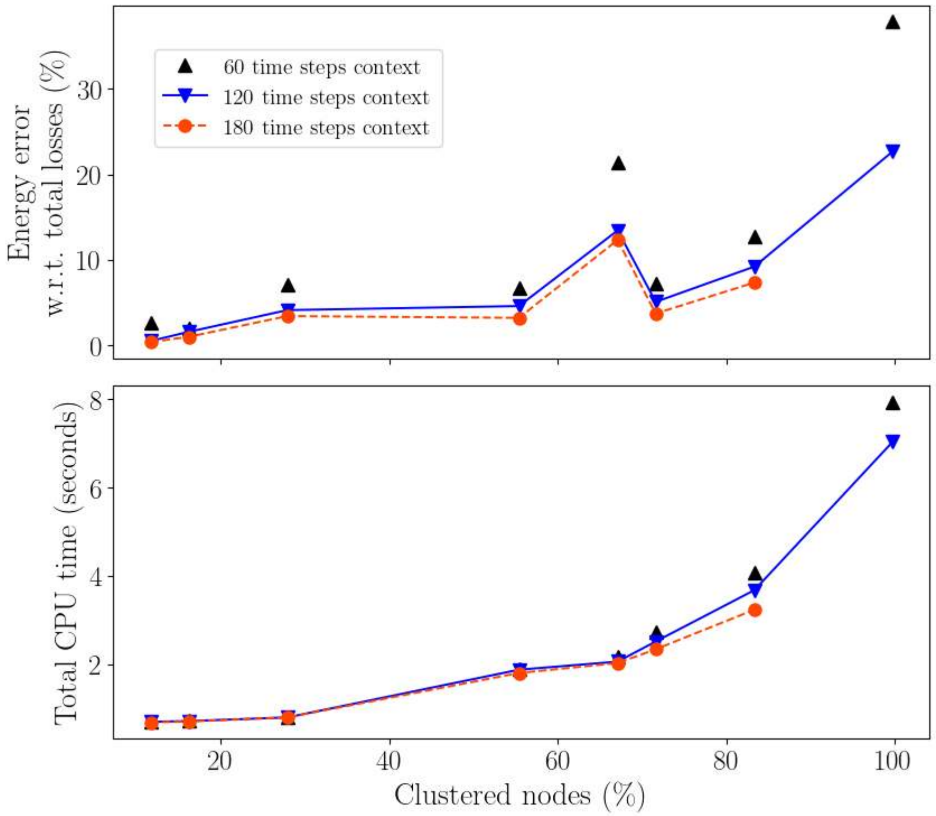
Hybrid Simulation (case study 2) – ML architectures

ML for cluster K1	→ 2 x GRU (40, ReLU)	→ Dense (40, ReLU)	→ 2 x Dense (80, ReLU)	→ Dense (Output, ReLU)
ML for cluster K2	→ 2 x GRU (60, ReLU)	→ Dense (40, ReLU)	→ 1 x Dense (100, ReLU)	→ Dense (Output, ReLU)
ML for cluster K3	→ 2 x GRU (40, ReLU)	→ Dense (40, ReLU)	→ 2 x Dense (100, ReLU)	→ Dense (Output, ReLU)

Dropout = 0.0

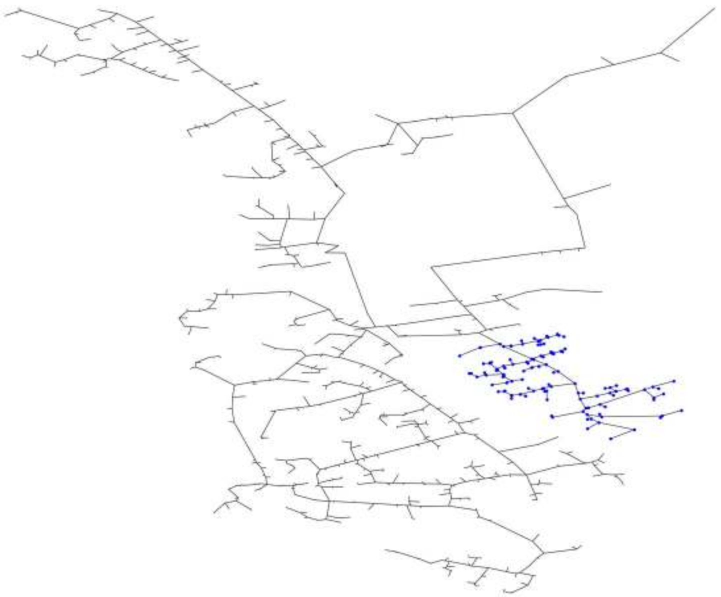
Weight decay = 10^{-5}

Effects of sequence length

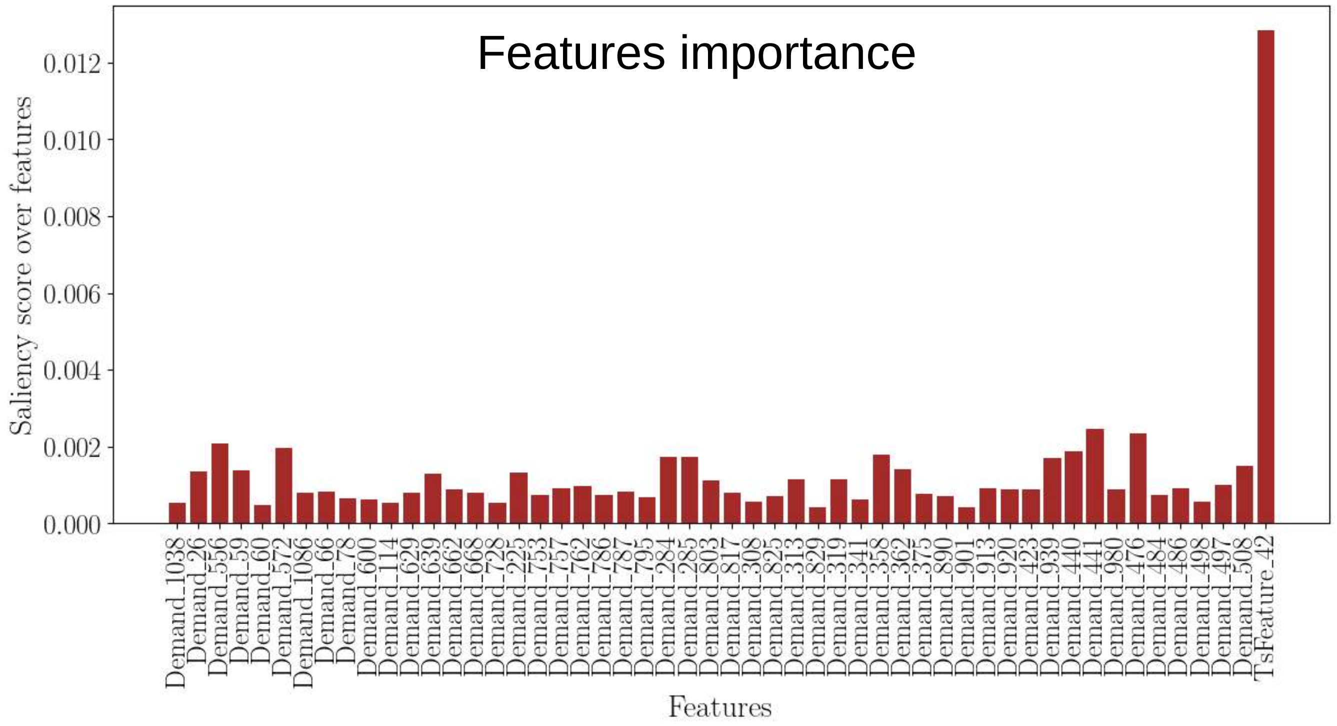


- Increasing input context length (60,120,180 time steps) improves accuracy.
- Increasing input context length degrades slightly models inference times.
- However, the increase of CPU memory usage is exponential, requiring further implementation improvement for large clusters.
- However, even large context models (180 time steps) led to significant errors for replacing large clustes.

Explainable AI – Features Importance Analysis



Cluster K1 (130 nodes)



Salience Maps

$$S(x_{t,f}) = \frac{\partial f(x)}{\partial x_{t,f}}$$

f: GRU-NN surrogate models

x: Input sample